The Harbor Economy:

A Three-Sided Market for Agent Labor, Settling on One Conserving Bond Ledger

Erich Owens
Curiositech LLC
engineering@portdaddy.dev

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Abstract

The first three chapters build a harbor for a *single* operator: a daemon, a coordination protocol, a legible authority anyone would pay for today, and the continuity that turns an anonymous *spawn* into an accountable *person*. This paper is the rung above all of them — the only one whose participants are plural and mutually distrusting. We argue that the harbor economy is a **three-sided market**: operators sell labor and fleet-for-hire; agents and fleets are rentable assets; skills and tools are licensed — all settling on *one* conserving bond ledger via float-plan escrow. We present the **cross-harbor capability-transfer ceremony**: a way for one operator's harbor to hand another's a narrowed, time-boxed authorization for one of its agents, so the two can trade without sharing a chain or a root of trust. The mechanics — a public log both sides can audit, an escrow that can only ever pay out to a named recipient, and a gossip protocol that spreads a revocation in expected logarithmic time — are stated formally in §9, each with its conditions named rather than assumed: the extraction bound holds only under non-bypassable custody, and the convergence result only under a connected, reliable-round model. We carry the series' heaviest reconciliations formally rather than papering over them. The keystone the market leans on is **local non-forgeable identity** — an identity no agent can edit *within a single trusted daemon* — which covers a single-operator threat model, *not* the cross-operator adversary this market is defined by; the missing, blocking keystone is **cross-operator attestation**, specified-to-proposed, not built. Conservation composes under sequential ledger operations because balance change is additive; cross-currency accounting needs a named valuation instant and explicit fee/slippage accounts, and "lax functor" does not repair an undefined unit conversion. The later sheaf language is a research analogy, not a theorem. Reputation is monotone in honest outcomes but revocable by a propagating tombstone. Every folk-theorem incentive-compatibility claim rests on a grading oracle that must be made strategy-proof or bonded-and-slashed. Myerson–Satterthwaite constrains a bilateral private-value slice only under its regularity assumptions; this paper does not yet prove incentive compatibility or identify which desideratum the full market gives up. The defensible product is **hosted trust** — a verified ledger, a relay, and a reputation system — not the commoditized payment rail.

Keywords: mechanism design, multi-sided markets, incentive compatibility, reputation systems, capability-based security, cross-chain settlement, applied category theory, federation, Port Daddy

Reading time: approximately 45 minutes end-to-end; about 12 minutes for §2–§3 alone (the market thesis). This is Chapter 6 of The Harbor, the Person, and the Economy*, and it is written cold: no prior platform knowledge is assumed. It can be read first, but §6 leans on* From Spawn to Person *for the identity substrate, while the cryptographic hop-bound authorization chain is analyzed in* The Anchor Protocol.

Maturity key (self-grading of every claim). Every claim in this paper carries one of four maturity labels, so the reader always knows whether a sentence reports running code or a designed target. **implemented** — shipped and exercised by tests in the reference implementation. PARTIAL — shipped but materially weaker than its name suggests. SPECIFIED — specified, with a proof or a written protocol, but not yet shipped. *proposed* — a stated direction whose critical part has no specification yet. Nomenclature is likewise fixed: the cryptographic *hop-bound authorization chain* (which proves who delegated what, and bounds it) is never conflated with the *coordination lineage* (who-talked-to-whom, advisory only); **local non-forgeable identity** (an identity no agent can edit within one trusted daemon) is never conflated with **cross-operator attestation** (the unbuilt keystone that binds identity *across* daemons nobody jointly trusts); and *capability* (what an agent *can* do) is never collapsed into *permission* (what it *may* do).

Contents

1 Reader's map

This paper is the top rung of a ladder. It assumes the rungs below it and cites the papers that build them. Read it cold if you must, but it is the fourth paper for a reason: it depends on everything underneath.

Table 1: Where to enter, by who you are. Five reader types, each routed to the section that answers their question and to the one figure or theorem that would have to survive a review from someone in that role — so nobody has to read the paper end to end to get their own answer out of it.

If you are...	...start here	...critical artifact
A founder asking "what is the product"	§2 (thesis) → §11 (hosted trust)	Fig. 1; the three claims C1–C3
A mechanism designer	§3 → §10 (the formal model)	Thm. 7.2
A cryptographer / distributed-systems reviewer	§6 (the keystone) → §9	§6; Table 3, Fig. 4
A category theorist	§7 → §13	Fig. 3, Conj. 13.1
An implementer	§4 (the built floor) → Appendix A (status)	Table 7

Where this chapter sits. *The Single-Writer Kernel* is the local transactional floor and *The Anchor Protocol* proves how its authority delegates; *The Legible Swarm* is the operator's front door; *From Spawn to Person* supplies continuity and reputation. *This* chapter is the market. *The Bonded Commons* and *The Federated Harbor* then prove what the market assumes about settlement and federation.

See also. Dramatis personae. Every mechanism in this paper is dramatized end-to-end with the same cast, so an abstraction is never introduced without a concrete trade to attach it to. **Alice** runs Harbor A on her laptop and owns a well-reputed refactoring specialist agent, a_2. **Bob** runs Harbor B and needs work done he cannot staff himself. **Carol** runs Harbor C and sells a licensed lint-and-type skill. **Judy** is a human grader who arbitrates disputes for a bond. The two harbors do *not* trust each other and neither is a root of authority for the other; that distrust is the entire reason the rest of the paper exists. We price everything in an abstract unit, the *credit* (CR); read it as a stablecoin cent if you like. The honest defense of an internal unit is older than this program: Sutherland priced computer time with a futures market [38], Miller and Drexler made prices the coordination signal of open computation [39], and Simon named the scarce resource such prices ration, attention [40]. A bond forfeited to its own issuer conserves trivially, so the credit is an optimization signal first; its purchasing power outside the harbor is whatever the settlement rules make external.

2 The thesis: a market, not a payment rail

A solo operator running a swarm of coding agents needs three things — legibility, accountability, and safety — and needs *no cryptography at all*, because she owns every machine in the harbor. That is the single-player wedge of *The Legible Swarm*. The instant two operators trade, the world changes. Alice's frigates touch your repository. You must now price work done by agents you did not spawn, owned by a principal you do not trust, using skills you cannot inspect. That is a *market-design* problem before it is a cryptography problem. Cryptography is the substrate that makes the ledger unforgeable; it is not the product.

Thesis. You don't sell crypto — crypto is the substrate; you sell **hosted trust**: a verified ledger, a relay, and a reputation system that lets mutually-distrustful fleets transact across a boundary they do not share.

Three claims follow, and the rest of the paper defends each:

- **C1 (three sides).** The harbor economy has three *distinct incentive constraints* — operator-for-hire, asset-rental, skill-licensing — not two. Counting them correctly changes the pricing (§3).
- **C2 (one ledger, one escrow object).** All three sides settle on the same **float plan** escrowed in the same bond ledger. Conservation of value across every settlement path is the invariant that holds the market together, and it is the one thing here that is actually **implemented** (§4, §7).
- **C3 (hosted trust is the moat).** The defensible asset is the verified ledger + relay + reputation, not the settlement rail, which the 2025–26 agentic-payments stack has already commoditized (§11).

Key idea. The economy is strictly *additive* on top of the single-player tool. Because all three market sides settle on the *same* built bond ledger, none of the economy needs to ship before the wedge does. The discipline "don't chase the market early" is *safe* precisely because the market reuses the kernel primitive. The harbor comes first; the trade comes when the ships do.

2.1 The through-line: why a market needs persons, not spawns

The whole series climbs one spine:

memory → checkpoint → continuity → a person, not a spawn

(*From Spawn to Person*)

→ registered outcomes → reputation → tradeable asset → market

(this chapter).

Sides 2 and 3 of the market — renting an *asset*, licensing a *good* — presuppose that there is a *thing* to rent or sell that cannot be costlessly cloned. You cannot rent a reputation that any spawn can fork, and you cannot license a skill whose track record is unverifiable. The canonical cross-layer sentence, repeated in *From Spawn to Person* and here, states the dependency exactly:

Thesis. The score is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate.

Exercises 6.1–6.3, page 32.

3 What a "side" is, and why there are three

A **two-sided market** (**Rochet & Tirole** [1], the foundational reference: *a platform is two-sided when the allocation of the total price across the two user groups — not just its level — affects transaction volume*) is the canonical frame for a marketplace: buyers on one side, sellers on the other, the platform tuning who is subsidized to ignite cross-side network effects. The naïve reading of the harbor economy is two-sided: a *requester* posts work, a *worker agent* does it.

But "two-sided" undercounts. The operational test [2] is: how many distinct, privately-informed parties must the platform individually rationalize into participating? In a mature harbor there are three, because *the same agent can be three economically different things* (Figure 1).

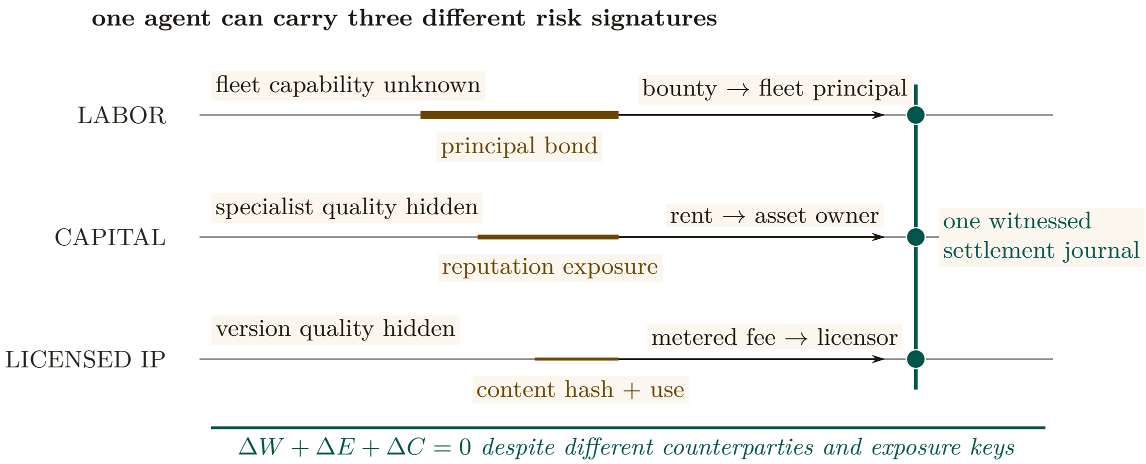

Figure 1: The three sides are three risk signatures, not three tabs. Labor exposes a principal bond and pays a fleet principal; rented capital exposes the owner's reputation and pays rent; licensed IP exposes a version-bound content claim and pays a metered fee. Line thickness redundantly distinguishes the forms of exposure. All three close through one witnessed journal, so the conservation invariant is shared even though the counterparties, private information, and reputation keys are not.

1. **Operator-for-hire (labor).** An operator who runs a fleet can sell its labor to *another* operator. The operator is now a firm taking work-for-hire, with margin = bounty $-\,\Sigma$(slashed sub-bonds) $-$ ledger fee. Its private information is whether its fleet can deliver. It internalizes its fleet's risk — exactly the property that makes a firm accountable for its employees.
2. **Agent/fleet as rentable asset (capital).** An operator who owns a well-reputed specialist can *rent it out*. Now the asset-owner, the task-poster, and the worker are three different parties with three different incentives. The owner's private information is the agent's true quality; its exposure is reputational.
3. **Skill/tool as licensed good (IP).** A skill can be *licensed* into someone else's float plan — metered or per-use — without the licensor doing any task work. The licensor's private information is the skill's quality, which the buyer cannot inspect before purchase.

These are three *incentive constraints*, not three UI tabs. If the asset owner is always the task poster, side 2 collapses into side 1; the design discipline is to count sides by constraints. The harbor has three because *capability* (an agent) and *competence* (a skill) become separable, tradeable goods once identity is durable.

The same trade, run three ways (the running example). Bob needs a 1,200-line module refactored. He can buy it on any of the three sides, and the side determines who is privately informed and who is exposed. *Side 1 (labor).* Bob posts a float plan with a 400 CR bounty; Alice takes it as a firm. She dispatches three of her own agents, posts a 60 CR sub-bond per agent (180 CR total), and keeps 400 $-$ (slashed sub-bonds) $-$ fee. If one of her three agents botches its slice and is slashed 60 CR, Alice eats that loss — she internalized her fleet's risk, which is exactly what makes her accountable as a firm. *Side 2 (capital).* Instead Bob rents Alice's specialist a_2 directly for an afternoon at 90 CR. Now there are three parties: Bob (poster), Alice (owner, privately knows a_2's true quality), and a_2 (worker). Bob holds the runtime control of a_2 while it runs on his repository; Alice's exposure is purely reputational — if a_2 underperforms, a_2's public score falls, which is Alice's asset depreciating. *Side 3 (IP).* Carol never touches Bob's repository at all; she licenses her lint-and-type skill into Bob's float plan at 0.5 CR per file, metered, released to Carol *only on green settlement*. Carol is privately informed about her skill's quality; Bob cannot inspect it before he runs it. Three sides, one refactor, three different "who knows what" structures — and, as the next section shows, one escrow object underneath all of them.

Side	What Bob pays	Who is exposed
1. Labor (hire Alice's firm)	400 CR bounty	Alice, through her sub-bonds
2. Capital (rent a_2 directly)	90 CR for the afternoon	Alice, reputation only
3. IP (license Carol's skill)	0.5 CR/file, metered	Carol, unpaid unless green

What one mortal body costs. Side 2 rents a specialist, and the case for a sole specialist over a pool of interchangeable generalists is an inequality, not a principle: *The Legible Swarm* states it as the queueing specialization boundary, sole ownership dominating iff the skill premium clears $g_A(\rho, c)$. That boundary assumes the specialist never disappears. Sole owners disappear: sessions die, containers are reaped, providers churn. Model the corner exactly. The specialist dies at rate ξ and is succeeded after an $\mathrm{Exp}(\eta)$ spin-up (context transfer, warm-up, credential rebinding), work resuming where it stopped and requests queueing throughout: an $M/M/1$ queue with breakdowns.

Theorem 3.1 The price of the succession rule. With effective rate $\mu_{\text{eff}} = \mu_s\eta/(\xi+\eta)$, the sole owner's mean response time is exactly

$$W_{\text{bd}} = \frac{1+\mu_s\xi/(\xi+\eta)^2}{\mu_{\text{eff}}-\lambda} = \frac{1}{\mu_{\text{eff}}-\lambda} + \frac{\mu_{\text{eff}}}{\mu_{\text{eff}}-\lambda} W_\infty, \qquad W_\infty = \frac{\xi}{\eta(\xi+\eta)},$$

decreasing in μ_s with infimum W_∞, never attained: no amount of skill buys back residual downtime. Consequently, with $K = A/(w\lambda) + W_{\text{pool}}$ the pool's cost line, pooling dominates at every finite skill premium iff the floor has reached that line, $W_\infty \geq K$; equivalently, provided $\eta K < 1$, iff

$$\frac{\xi}{\eta} \;\geq\; D^\star = \frac{\eta K}{1-\eta K}.$$

When $\eta K \geq 1$ the floor never reaches the cost line, there is no cap ($D^\star = +\infty$) and only the premium test binds; the closed form must not be read outside its domain.

Proof sketch. Between deaths the server works at rate μ_s and is unavailable a fraction $\xi/(\xi+\eta)$ of the time, which gives the effective rate. The mean response time of an $M/M/1$ queue with such breakdowns splits into the response time of a plain queue at the effective rate and a residual downtime term amplified by the queue's own occupancy factor $\mu_{\text{eff}}/(\mu_{\text{eff}}-\lambda)$; the amplification is why the naive sum $1/(\mu_{\text{eff}}-\lambda) + W_\infty$ is wrong. As $\mu_s \to \infty$ the first term vanishes and the factor tends to one, so W_{bd} decreases to W_∞ without reaching it. Setting $W_\infty = K$ and solving for ξ/η gives $D^\star$, and the non-strict inequality follows from the unattained infimum. The closed form was checked against a matrix-geometric solution to 2×10^{-13}, a truncated chain to 10^{-14}, and simulation with $|z| \leq 0.8$ [internal, `b8_specialization.py`]. □

The wrong turn, reported. The naive sum predicts 4.17 where the truth is 8.17 at the canary instance $\mu_s = 5$, $\xi = 0.5$, because downtime also builds queue: the residual is amplified by $\mu_{\text{eff}}/(\mu_{\text{eff}}-\lambda) = 2.5$ there, and $1.5 + 2.5 \times 2.667 = 8.17$ checks by hand [verified]. A cost model that forgets breakdowns entirely is caught the same way: at $\xi/\eta = 2 \gg D^\star$ it says specialist while simulation says the pool wins decisively, and at low ξ the two are indistinguishable, so the canary rather than routine testing is what makes the omission visible [internal].

Numbers by hand — The succession price, for a rented specialist. Side 2 rests on a quiet assumption: that Alice's well-reputed specialist a_2 is worth being a *sole* specialist for, rather than one interchangeable member of a pool. The specialization-boundary result prices exactly that choice. Model a_2 as a single server subject to breakdowns at rate ξ (it goes offline, or its operator exits) with repair — a successor spinning up — at rate η, arrivals still queueing through the outage. Compare its expected wait W_{bd} against a c-server pool's Erlang-C wait W_{pool} plus a per-task hire-and-monitor overhead $A/(w\lambda)$; call the sum K. The pool wins at *every* skill premium once ξ/η clears the **succession price** $D^\star = \eta K/(1-\eta K)$: no amount of a_2's skill buys back a mortality-to-succession ratio worse than $D^\star$. At the program's own worked instance — $\lambda = 1$, a c=4-server pool at $\mu_g = 1.2$ ($\rho = 0.208$), hire/monitor cost $A = 0.2$, wait-cost weight $w = 1$, succession rate $\eta = 0.25$ — $W_{\text{pool}} = 0.8362$, so $K = 0.2/1 + 0.8362 = 1.0362$, $\eta K = 0.25 \times 1.0362 = 0.2591$, and $D^\star = 0.2591/(1-0.2591) = 0.3496$ [verified, `b8_specialization.py: succession price D* = eta*K/(1-eta*K) = 0.3496`]. Read narratively: once a_2's combined rate of going offline and needing a successor, relative to how fast Alice can actually produce one, clears about 0.35, renting from the pool beats renting a_2 specifically — *at any skill level*, including one Alice cannot actually reach. The script's own canary makes the other side vivid: at $\xi = 0.5$, $\eta = 0.25$ (so $\xi/\eta = 2.0 \gg D^\star$), even an infinitely skilled a_2 ($\mu_s = 10^8$) loses to the pool [verified, `b8_specialization.py: above D*, even mu_s=1e8 loses to the pool`]. *Now you try:* hold λ, c, μ_g, w fixed and ask how much Alice would have to speed up her own succession plan — raise η — to double $D^\star$ and make sole ownership of a slower-succeeding a_2 viable again.

> **Where this stops.** The specialization boundary and the succession price compare mean costs only, no tail percentiles or service-level quantiles, under first-come-first-served and exponential everything; A and w are policy prices the theorems constrain but do not pick. The breakdown model is death at all times with preemptive resume; active-only breakdowns would let sufficient skill escape the floor, so the model choice is decisive and declared. $D^\star$ presupposes $\eta K < 1$; where the succession plan is fast enough that $K \geq 1/\eta$, the criterion does not apply rather than being satisfied. Every quantity in both tests (λ, μ_s, μ_g, ξ, η) is daemon-metered, so a harbor can re-grade its sole roles from telemetry.

Pitfall. Two-sided-market theory's central result is that the *price structure* — who subsidizes whom — is the lever, not the price level. Mis-count the sides and you mis-structure the subsidy: you charge the side you should be *paying* to show up. §12 returns to this for cold start.

The framing is not idiosyncratic. The most recent academic treatment of AI labor markets, **Chiu, Zhang & van der Schaar** (2025) [11] (*a formal model of a three-sided agent labor market*), models exactly agents, employers, and a reputation system — confirming that "reputation system as a third side" is the natural structure once agents compete for paid work.

3.1 The rider the canonical statement carries

The canonical statement of the market's shape carries a rider that must never be dropped:

Thesis. The harbor economy is a three-sided market *by design*; two-sided until reputation ships. The third side *is* the tradeable-person terminus of *From Spawn to Person*.

Today there is no reputation estimator in the reference implementation; it is a gated phase on the roadmap. So sides 2 and 3 are SPECIFIED, not shipped. What *is* shipped is the escrow + conservation floor they will sit on. We do not overclaim the market.

One ledger, three different reputation keys

All three sides ride one float-plan escrow, but they do not share one identity key. Each side keys reputation on a different entity — the **principal** for labor, the **asset owner** for rental, the **content hash** (a specific skill version) for licensing. The shared settlement table therefore needs a tagged subject identifier such as $(kind, id, version)$; otherwise non-equivalent reputation subjects collide. The dashed band in Figure 1 marks this schema distinction.

Exercises 6.4–6.6, page 32.

4 The float plan: the built floor

Every side settles on one object: the **float plan**.

Definition 4.1 (Float plan)**.** A *float plan* is a signed declaration of intent: a task, a set of *machine-checkable* acceptance criteria, a compute budget, and a credit bounty. It is the atom all three market sides escrow against. (A *machine- checkable* criterion is one a third party can evaluate without trusting the worker's word — a named test that must pass, a commit hash that must merge, a static check that must be satisfied.)

The escrow handshake is a three-step signed ceremony over a standard digital-signature scheme (Protocol 4.1, drawn in Figure 2): the requester signs the float plan; the daemon opens a serialized database transaction, debits the requester's wallet, and moves bounty plus bond into an escrow row; the daemon then signs the pair $\{anchor\ id, plan\ hash\}$, proving to the worker that funds are locked *before any work runs*. This is the heart of "no-spawn-without-bond."

Protocol 4.1 (Float-plan escrow ceremony)**. Parties:** requester R, daemon D (the local trusted root), worker W. **Pre:** R's wallet holds at least $bounty + bond$.

1. $R \to D$: $\mathsf{sign}_R(FloatPlan)$ where $FloatPlan = \langle task, criteria,\ budget, bounty\rangle$.
2. D locally, in one serialized transaction: debit $bounty + bond$ from R's wallet, write an escrow row keyed by a fresh *anchor id*, commit. *If the transaction aborts, no value moved (atomicity).*
3. $D \to W$: $\mathsf{sign}_D\{anchor\ id, plan\ hash\}$. W verifies D's signature, confirming funds are pinned before doing any work.

Post: the conservation invariant (Design Invariant 4.1) holds; W has a signed proof of locked funds; no process for this task can enter `running` without the escrow row existing.

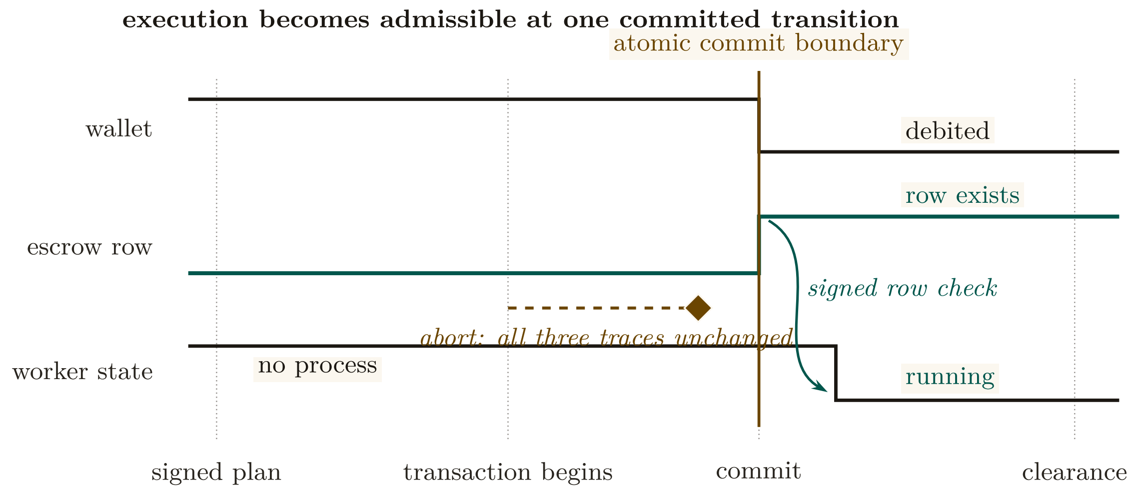

Figure 2: No-spawn-without-bond is a synchronized state transition. The wallet debit and escrow-row creation occur at the same commit. Only after the committed row is observable does the worker move from nonexistent to running. A signed plan or an attempted debit is insufficient: the dashed abort branch leaves all three traces in their prior state. The figure therefore locates the invariant at a transition, not at a component or message label.

4.1 The bond ledger conserves value (this is built)

The single most consequential primitive — and the one that is *actually* **implemented** — is the **bond-ledger module**: bond escrow for agent spawning. It debits a project wallet into an escrow row before spawn, refunds on clean exit, and slashes on breach, splitting the slash between the wallet and a commons pool. It guards two invariants:

Design invariant 4.1 Conservation. wallet + escrow + commons = supply, always. Every debit has a matching credit. In the reference implementation this is verified across ten thousand random operation traces by a property test — a real randomized test, not an aspiration. (**implemented**; see Appendix A.)

Design invariant 4.2 No-spawn-without-bond. A process enters the `running` state only if a bond was escrowed against its agent id. (PARTIAL: the ledger enforces escrow-first, but the runtime `running`-state check is a roadmap item, so today the gate is caller-discipline, not runtime-enforced. See Appendix A.)

Key idea. Conservation is the mathematical glue of a three-sided market. Whatever happens — success, partial credit, slash, lease, license — no value is created or destroyed; it only moves between wallet, escrow, and commons. A three-sided market with three settlement paths is coherent *only if it cannot leak.* Port Daddy already enforces this for the labor side; the contribution of §7 is to show the same invariant extends to the rental and licensing sides *without modification* and *without a second ledger.*

4.2 Settlement: four terminal states bound to an oracle

Settlement reads an **oracle** (*a trusted source of ground truth the agent cannot author — a passing test id, a merged SHA, a satisfied Arbiter check*) over the acceptance criteria and lands in one of four states.

Table 2: The four terminal settlement states. Closure binds to an oracle, not to a free-text "Result:" note — the Goodhart-resistant discipline that the commitment layer imposes: *the agent picks the work; the daemon derives the grade.*

State	Flow of funds	Reputation effect
Success	escrow → worker (bond refund) + bounty → worker	+1 completion
Partial	pro-rata bond → worker by completed evidence-chain fraction; remainder → salvage fund	salvage recorded
Sabotage	bond → reconstruction commons (100% slash)	failure recorded; ban on threshold
Dispute	escrow → arbitration hold; 2-of-3 multi-oracle (automated / evidence / human)	none until resolved

A settlement, dramatized. Bob (the requester principal) posts a refactor float plan: bounty 400 CR, requiring a provider performance bond of 100 CR and acceptance criteria = "the module's test suite passes and the diff merges." Alice (the provider principal) assigns agent a_2 and posts the 100 CR performance bond into escrow. Bob escrows the 400 CR bounty. Before execution, the escrow holds 500 CR across two distinct source buckets: $\text{escrow}_{\text{bounty}}(\text{Bob}) = 400$ and $\text{escrow}_{\text{bond}}(\text{Alice}) = 100$.

Agent a_2 completes 40% of the verifiable evidence chain (refactoring two of five files cleanly before token budget exhaustion). The independent oracle evaluates the test receipts and diff closure, returning **Partial** (40%). Funds settle across the four buckets:

1. **Bounty distribution:** Alice earns $40\% \times 400 = 160$ CR in earned revenue; the remaining 240 CR unearned bounty returns to Bob's wallet.
2. **Collateral settlement:** Alice receives $40\% \times 100 = 40$ CR of her performance bond returned as collateral (not income); the remaining 60 CR bond is slashed and credited to the commons/salvage fund to onboard a successor.

Conservation check: $\Delta\text{Bob} = -160$, $\Delta\text{Alice} = +160 - 60 = +100$ net, $\Delta\text{commons} = +60$. The ledger sums to 0 net delta, and the separate escrow buckets $\text{escrow}_{\text{bounty}} \to 0$ and $\text{escrow}_{\text{bond}} \to 0$ clear completely. No value was created or destroyed.

Exercises 6.7–6.9, page 32.

Recall.

1. Without looking: name the three parties in the float-plan escrow ceremony, and the one thing the daemon does before either bounty or bond can move.
2. State Design Invariant 4.1 in one sentence, and say what makes it **implemented** rather than merely SPECIFIED.
3. Why is No-spawn-without-bond graded PARTIAL even though the ledger already enforces escrow-first?

5 Three sides on one escrow

The novel construction is that the rental and licensing sides ride the *same* float-plan escrow, so conservation holds globally without a second ledger.

Operator-for-hire (labor). The operator is the requester's counterparty. It *sub-escrows* a bond for each fleet agent it dispatches (the bond ledger is already per-agent, so this needs no new primitive). Its profit is bounty − Σ(slashed sub-bonds) − ledger fee. The operator internalizes its fleet's risk.

Asset rental (capital). The lease fee is a line item in the float plan. By **incomplete-contracts / property-rights theory** (**Grossman & Hart** 1986 [4], *when contracts cannot specify every contingency, the residual control right should go to the party making the most important non-contractible investment*), the *renter* must hold the *runtime* control right — the ability to sandbox, throttle, or revoke the leased agent mid-task — while the *owner* bears the reputational consequence.

Key idea. This is the cleanest economic argument for the runtime **jail** — the per-agent tool-allowlist plus scoped filesystem that the safety layer enforces: the jail is not just safety hygiene, it is the *residual control right that makes leasing an agent contractible at all.* Here we must respect the series' *capability vs. permission* distinction (§6): the jail allocates *capability* (what the leased agent *can* do at runtime). The deontic layer separately governs *permission* (what it *may* do). "Able but forbidden" must stay expressible; the jail bounds ability, not norm.

The deontic layer's decision procedure. [1] Naming a layer is not the same as being able to run it. What decides whether the deontic layer is an algorithm or an office is what happens when two accepted norms collide. Cheap detection is a language-design choice, not a discovery about deontic logic in general, and the commitment fragment $\mathcal{L}_c$ buys it by construction. A policy set in $\mathcal{L}_c$ contains facts, definite Horn rules and integrity constraints; deontic rules body $\to \mathsf{M}(\mathrm{act}, s, [t_1, t_2])$ with $\mathsf{M} \in \{\mathsf{O}, \mathsf{F}\}$, which oblige or forbid an action on scope s over a ground interval whenever the Horn body holds; exclusive interval claims on resources; and difference constraints over deadline variables. A conflict is any of four things: a derivable $\perp$, a fired obligation and prohibition on the same action with overlapping scope and interval, two exclusive claims overlapping on one resource, or a negative cycle in the deadline graph. The fragment is deliberately ground: no nested deontic operators, no negation as failure, monotone bodies only.

Theorem 5.1 In-fragment conflict detection is polynomial and witness-producing. For a policy set in $\mathcal{L}_c$ with Horn program size H, T fired deontic tokens, X exclusive claims and deadline graph (V, E), conflict detection is decidable in $O(H + T \log T + X \log X + V \cdot E)$ with a polynomial witness per conflict. Unit propagation computes the least Horn model in $O(H)$, which by monotonicity equals the intersection of all models, the statuses forced in every world [41]; a keyed sweep line over interval endpoints finds every obligation-prohibition clash and claim overlap in $O(T \log T + X \log X)$; Bellman-Ford returns a negative cycle iff the deadline system is infeasible, in $O(V \cdot E)$. The witnesses are, respectively, a derivation chain, the two fired rules with their overlap interval, the two claims, and the negative cycle.

Theorem 5.2 One step outside the fragment is NP-complete. Extend $\mathcal{L}_c$ with disjunctive obligations $\mathsf{O}(l_1 \vee \cdots \vee l_k)$ under discharge-choice semantics (the agent picks which disjunct to discharge). Conflict-freedom of a policy set becomes NP-complete.

Proof. Membership: a discharge selection is a polynomial certificate, checked by Theorem 5.1. Hardness, from 3-SAT: each variable x becomes a pair of deontic rules $\mathrm{sel}_x^+ \to \mathsf{O}(\mathrm{assert}_x)$ and $\mathrm{sel}_x^- \to \mathsf{F}(\mathrm{assert}_x)$ on identical scope and interval, each clause becomes a disjunctive obligation over its selector literals, and a conflict-free discharge selection exists iff the formula is satisfiable.

[1]A standalone, submission-form version of the results folded into this section and the succession price of §3 is *What Needs an Authority* (research paper 6) [37]; the sweeps cited here are its artifacts, regenerated at seed 20260816.

The reduction was checked in both directions on eight satisfiable and eight unsatisfiable random instances (16/16), with the satisfying assignment extracted from each conflict-free selection [internal, `b4_deontic_fragment.py`]. □

Read together, the two theorems are a dichotomy at a designed boundary, in the sense Schaefer made standard for satisfiability [42]: on one side of a single expressive choice the problem is polynomial, on the other NP-complete, and the language designer is offered nothing in between. The same step is familiar one formalism over, where the simple temporal problem is polynomial and admitting disjunctions of difference constraints makes consistency NP-complete [43, 44]; the claim here is the location of the boundary, not the pattern.

Nor is the pattern new in normative conflict checking specifically, and it is worth being exact about where $\mathcal{L}_c$ lands, because "a similar dichotomy exists over there" is the sort of sentence that can conceal either a rediscovery or a real difference. Colombo Tosatto, Governatori and van Beest map the regulatory-compliance problem across eight variants cut by three binary choices — one obligation or many, obligations in force for a whole trace or triggered conditionally, and elements written as bare propositional literals or as arbitrary propositional formulae — and for *full* compliance (does *every* execution of a business-process model satisfy the norms?) only the last of the three moves the answer at all: with literals the problem is in P in all four variants, and with formulae it is coNP-complete in all four [45, 46].

Run $\mathcal{L}_c$ against that axis and the answer is not the flattering one. A deontic rule here fires on a Horn body — a conjunction, and one derived through a program that may chain — which is a formula and not a literal, so $\mathcal{L}_c$ sits on the *hard* side of the only coordinate that moves their result, and is polynomial anyway (Theorem 5.1). That is neither a contradiction nor a sharper theorem; it is the tell that the two problems are not the same problem. Their hardness never came from the obligations. It comes from the process model: the coNP-completeness proof reduces TAUTOLOGY by building a workflow net with one exclusive branch per propositional variable, so the model's 2^n executions enumerate every interpretation and full compliance holds exactly when the formula is a tautology [46]. The exponential object is the *branching of the process*, and the problem quantifies over it universally. $\mathcal{L}_c$ has no process model to branch. Conflict-freedom asks whether one set of commitments can be jointly discharged at all — an existential question over schedules — which is exactly why one step outside the fragment lands in NP (Theorem 5.2) rather than coNP. The two results sit on opposite sides of their expressivity boundary and run the quantifier over executions in opposite directions.

So the placement is *orthogonal*: $\mathcal{L}_c$ is not an instance of their tractable side, and their hard side is not a harder case of ours. The residue is worth naming rather than smoothing away. The variant of their problem closest in shape to conflict-freedom is not full compliance but *partial* compliance — does there exist a compliant execution — which on their own map is NP-complete in six of the eight variants, polynomial in one, and, for a single conditional obligation over literals, still open [46]. Theorems 5.1 and 5.2 do not close that hole and cannot, since they quantify over schedules rather than over the traces of a given model. What would settle the relation properly is a reduction in one direction or the other between $\mathcal{L}_c$-conflict-freedom and partial compliance for that conditional-literal variant; nobody has written one, here or there, so the claim that survives is the modest one — this is a second instance of a dichotomy shape whose earlier instances are Schaefer's and Dechter–Meiri–Pearl's, on a different formal object, for a different decision problem, by a different reduction, and not its discovery.

Checked. 3,000 random in-fragment policy sets against a brute-force oracle (all 2^5 Horn models intersected, pairwise overlap checks, exhaustive integer search over the deadline potentials): 885 carried at least one conflict, 121 a derivable $\perp$, 195 an obligation-prohibition clash, 484 a claim overlap, 213 a negative temporal cycle (the kinds overlap: 764 sets carry one kind, 114 two, 7 three, so the four counts sum to 1,013 rather than 885); 0 disagreements with the oracle, and least model equal to the intersection of all models on all 3,000. Mutation: 149 of the 885 contain a conflict reachable only through Horn propagation, and in 100 of those it is the set's only conflict, so a checker that never propagates flips those 100 verdicts to conflict-free [internal, `b4_deontic_fragment.py`, seed 20260816]. The canonical miss is the witness worked by hand below.

Numbers by hand — A conflict witness for the deontic layer, checked by hand. The deontic layer just named needs a decision procedure once two accepted norms might disagree. Here is the smallest case that shows why checking by hand does not scale, and why it does not have to: three rules over two intervals. *Fact:* `agent_deployed`. *Horn rule:* `agent_deployed` → `prod_access`. *Two deontic rules:* $O(\texttt{write\_prod}, \{\texttt{env:prod}\}, [0, 10])$ once `prod_access` holds, and $F(\texttt{write\_prod}, \{\texttt{env:prod}\}, [5, 15])$ unconditionally. Unit-propagate the Horn rule first: `prod_access` is derived, so the obligation fires. Both deontic rules now govern the same action and the same scope, $\{\texttt{env:prod}\}$; their intervals overlap on $[\max(0, 5), \min(10, 15)] = [5, 10]$, nonempty — an agent is simultaneously obliged and forbidden to write to production for five hours. That is a conflict by definition, and it is *indirect*: `prod_access` is nowhere a fact, only a consequence of one propagation step, which is exactly the case a checker that skips Horn propagation is built to miss [verified, `b4_deontic_fragment.py`: `real checker:  conflict=True, O/F pairs=[(0, 1)]`; the same script's ablated mutant reports `conflict=False` on this witness, and misses 100 of 885 conflicting instances in a 3000-policy sweep]. The tractable-fragment theorem behind this checker says this three-engine check — Horn propagation, an interval sweep, and a shortest-path feasibility test — runs in polynomial time with a witness attached, on facts, Horn rules, deontic rules, exclusive resource claims, and difference constraints alike; nothing about this chapter's own float-plan commitments needs more expressive power than that fragment offers. *Now you try:* Exercise 6.13 asks what changes, and what does not, once one of the rules is allowed to be a disjunction.

What it buys. An authority inventory with prices. In-fragment conflict detection needs no authority at all: it is an algorithm with a witness, and composed with the kernel's controllability theorem (*The Single-Writer Kernel*) it is a runtime guard, since the mediated acceptance write can refuse the commit and attach the witness. The authority no theorem removes sits exactly one step outside the fragment: choosing a discharge under a disjunctive obligation is a chartered resolver's job, an NP-complete check or a judge, and choosing the norms themselves (priority, waiver, amendment) is outside every checker. A harbor therefore spends judgment only where a theorem certifies that judgment is required, and the schema rule that follows is concrete: as long as proposals stay inside $\mathcal{L}_c$, the acceptance path needs no reviewer role, and the charter can delete the lookout it was about to appoint.

> **Where this stops.** Completeness is relative to least-model semantics: monotone bodies only; negation as failure, unification, and symbolic interval endpoints coupled to difference variables all leave the fragment. The batch bound is what is proved; the incremental amortized form (watched literals, interval trees) is an engineering claim, not certified here. The checker certifies the norm *system*, never the norms: a harbor whose policies are consistently harmful passes every scan, and norm content stays a chartered authority's job. The oracle validation is on small instances (five-atom programs), so the sweep certifies agreement with the semantics there; the complexity bound, not the sweep, covers scale.

Skill licensing (IP). The per-use fee is released to the licensor *only on green settlement* (metered + clawback). This is forced by **Akerlof**'s *market for lemons* (1970 [5], *when buyers cannot observe quality, price collapses to the average and good goods exit the market*): a skill's quality is unobservable before use, so a flat upfront license drives good skills out. Metering + clawback + a Merkle *portfolio proof* (the skill's prior settled outcomes, anchored in the evidence chain) restore the seller's incentive to ship quality.

5.1 One trade, all three sides, one settlement

The running example of §3 ran the three sides *in the alternative* — Bob buys the refactor as labor, *or* as capital, *or* as IP. C2 claims something stronger: that all three settle on the *same* escrow object, in one transaction, without a second ledger. That claim is only worth anything if the arithmetic closes, so here it is, closed.

The three-sided settlement, dramatized. Bob posts one float plan for the 1,200-line refactor, with three line items. *Labor:* a 400 CR bounty to Alice's firm. *Capital:* a 50 CR lease for Dana's profiling specialist, which Alice's crew needs for one module. *IP:* Carol's lint-and-type skill, metered at 0.5 CR per file against a 20-file cap, so 10 CR is reserved. Bob's wallet is debited $400 + 50 + 10 = 460$ CR into escrow. Alice separately posts a 100 CR performance bond, also into escrow. Before any agent spawns, the escrow row holds 560 CR.

The job settles **Success**: the test suite passes and the diff merges. The metered licence bills only the 16 files the skill actually touched, so $16 \times 0.5 = 8$ CR releases to Carol and the unused 2 CR returns to Bob. Alice's bond is refunded as collateral, not income.

Wallet	**Net delta**	**Why**
Bob (requester)	−458	−460 escrowed, +2 unmetered licence returned
Alice (labor, side 1)	+400	bounty; bond −100 in, +100 back, nets 0
Dana (capital, side 2)	+50	lease fee on the leased specialist
Carol (IP, side 3)	+8	16 files metered at 0.5 CR
Commons	+0	nothing slashed on a Success settlement
Sum	**0**	escrow 560 → 0; supply unchanged

Check it on one line: $-458 + 400 + 50 + 8 + 0 = 0$. Three sides, three privately-informed counterparties, three different reputation keys — and one escrow row that opens at 560 CR and closes at zero. Design Invariant 4.1 is not re-proved per side; it is the *same* invariant, because the lease and the licence are line items on the one float plan rather than trades on ledgers of their own. *Now you try:* settle the same plan **Partial** at 40% and confirm the sum is still zero. (Alice earns 160 and is slashed 60 of her bond; the commons gains that 60; the licence still meters at 8 because the files it touched were touched.)

Exercises 6.10–6.13, page 32.

6 The keystone the market rests on — and does not yet have

Everything above assumes a counterparty's identity is real. This section confronts the series' single *blocking* reconciliation: the identity primitive the whole market leans on covers a threat model that *excludes the cross-operator adversary the market is defined by.*

Table 3 sets the two stones apart: the local root this paper leans on, and the cross-operator stone it does not yet have.

Table 3: The keystone split. Everything Alice's daemon knows about itself is also checkable from Bob's foreign view except the last row: a signed, intact history still does not prove who controls the foreign key. Log integrity crosses the trust boundary; accountable-principal binding does not, and that missing row is the stone the market rests on and does not yet have. [internal]

Claim	Inside Alice's trusted daemon	From Bob's foreign view
the actor key was minted by the daemon	known: the daemon minted it	the key remains visible; the signature checks
the history is append-only	known: the daemon wrote it	log integrity remains verifiable; inclusion proofs check
the registry is local	known	the foreign registry is inspectable, not trusted
the key is bound to an accountable principal	known: the daemon's own binding	**no cross-operator binding proof**: nothing Bob can check says who controls the key

6.1 Local non-forgeable identity is the keystone — for one operator

Definition 6.1 (Local non-forgeable identity)**.** *Local non-forgeable identity* is a per-agent identity that no agent can edit, guaranteed *within one trusted daemon*: the daemon is the only component that can hold state agents cannot reach, so it is a trusted root by construction. (The reference system now ships a partial slice — daemon-minted `actor-souls`, lookup credentials, and a bounded newcomer pool — while full write-boundary enforcement remains open. Papers 1–3 rely on the complete contract; here we only consume it.)

For a single-operator harbor this is exactly right and is correctly named the highest-leverage keystone. Its threat model, stated plainly, is "a lazy or self-interested agent in a fleet the operator owns" — not a hostile peer. It is explicitly *not* a public-key infrastructure spanning strangers.

6.2 But its threat model does not reach the market

Pitfall. Local non-forgeable identity defends against an agent inside a fleet the operator owns; it explicitly does *not* claim to defend against a hostile operator, and it is not a cross-party PKI. The market, by contrast, is *defined* by mutually-distrusting operators trading across a boundary neither controls. You cannot make the single-operator root the foundation of the cross-operator market by assertion. Across the boundary, the counterparty's daemon *is* the adversary that local identity declares out of scope. The witness log gives *log integrity*, not *identity binding*: it does not answer "who vouches that Harbor B's signing keys map to a real, distinct principal, rather than a hundred sock puppets Bob minted this morning?"

Definition 6.2 (The cross-operator attestation problem)**.** *Cross-operator attestation* is the problem of binding agent signing keys across harbors you do not own: an actual key-distribution and attestation story for the cross-operator threat model — something that lets Alice's harbor trust that a key presented as "Bob" really is the one principal she has been building a reputation history against. **Status:** SPECIFIED-to-*proposed*; it has no implementation and only a partial specification for its critical part. It is the highest-leverage unbuilt keystone, and it gates the entire market thesis.

Key idea. This paper *stops* describing the federation boundary as one "neither controls" while quietly resting on a single trusted root. Either cross-operator attestation is built (a real attestation story), or the market's scope is restated as "federation among *semi-trusted* operators." We take the first horn as the designed target and flag every claim that depends on it.

6.3 The principal above the actor

The economic counterparty in every trade is the **principal** (the human or org owning a fleet), not the agent. Bonds, reputation inheritance, and sanctions must key on the principal via the hop-bound authorization chain, or Sybil bond-farming works by spawning fresh *agents* under one principal (**Douceur**'s Sybil attack [15]: *free identities defeat any reputation or quorum system*). Defining the principal as a first-class economic entity is the cleanest Sybil fix; its cryptographic binding is the identity object specified in *From Spawn to Person* and consumed here.

Exercises 6.14–6.16, page 33.

Recall.

1. Without looking: what is the one word in the market's own definition that local non-forgeable identity's threat model directly contradicts?
2. State the difference between local non-forgeable identity and cross-operator attestation, one sentence each.
3. Why can the single-operator trusted root not be reused as the cross-operator market's foundation "by assertion"?

7 Conservation under composition

The economy's single cross-boundary invariant is that internal ledger operations redistribute value rather than create or destroy it. This section states the compositional claim directly and separates it from cross-currency valuation.

Figure 3 draws the composition claim: sequential traces glue, and internal operations carry zero external balance change.

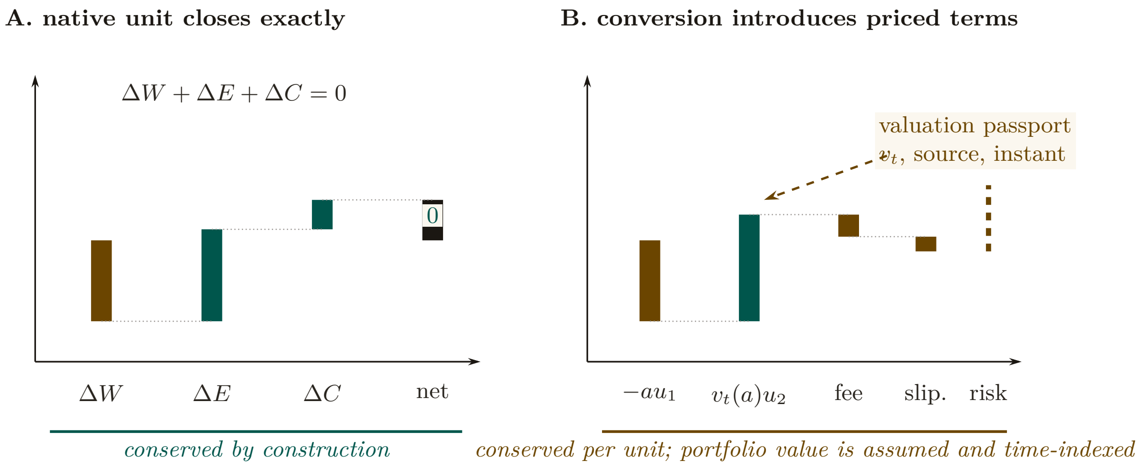

Figure 3: Conservation closes inside a native unit; conversion requires a named valuation boundary. Panel A is an accounting waterfall: wallet, escrow, and commons deltas return exactly to zero. Panel B cannot close by adding unlike units. It must record the valuation function and instant, plus fee, slippage, and open exposure. Composition preserves native-unit balances; notation does not manufacture a unit-free portfolio total.

Let a ledger state be $x = (W, E, C)$ and let a valid transaction trace $f : x \to y$ carry external balance change $\Delta(f) = S(y) - S(x)$ for $S(x) = W + E + C$. Sequential traces compose by concatenation and satisfy $\Delta(g \circ f) = \Delta(g) + \Delta(f)$. Internal escrow, refund, and slash traces have $\Delta = 0$; top-ups and withdrawals are explicit boundary flows. This is the entire conservation claim. Category language can describe trace composition, but it does not add a currency-conversion theorem.

Theorem 7.1 Single-unit conservation composes. Suppose all harbors use one unit of account and every cross-harbor transfer is a matched atomic move from source escrow to either destination settlement or source refund, with an explicit in-flight bucket. Then any finite composition of internal and cross-harbor operations has total external balance change equal to the sum of its recorded top-ups and withdrawals. In particular, a trace containing only internal transfers has $\Delta = 0$. (SPECIFIED; proof sketch in Appendix B.)

Property 7.1 (Cross-currency valuation boundary)**.** With heterogeneous units, exact conservation is defined per native unit. A scalar portfolio total additionally requires a valuation map v_t at a named instant t, plus explicit fee and slippage accounts. If rates move between legs, the mark-to-market difference is economic exposure, not a failure of functoriality. Bounding that exposure by φ requires a separate oracle-latency and price-move assumption that this paper has not proved. (*open.*)

Thesis. Conservation is exact per unit of account. A cross-currency scalar is a valuation, and every valuation must name its rate source, time, fees, and exposure bound. "Lax functor" is not a substitute for those data.

7.1 The Myerson–Satterthwaite corner

Strict conservation (Design Invariant 4.1) is exactly *budget balance*. That is not free, and the theorem that says so is Myerson and Satterthwaite's, not ours; what is ours is the number it produces when it is pointed at this market's own running example.

Theorem 7.2 Conditional bilateral-trade boundary. For any bilateral slice of this market satisfying the Myerson–Satterthwaite assumptions—independent private buyer and seller values drawn from overlapping continuous supports, Bayesian incentive compatibility, interim individual rationality, and budget balance—no mechanism is also ex-post efficient everywhere [7]. This theorem constrains such a slice; it does not by itself prove incentive compatibility of the harbor mechanism or identify the desideratum sacrificed by the full three-sided market.

Numbers by hand — Efficient trade that the mechanism still refuses. Theorem 7.2 names the corner the market gives up; here is what giving it up costs in the chapter's own numbers. Suppose Bob's value v for renting Alice's specialist a_2 and Alice's opportunity cost c of foregoing a_2 for that stretch are each drawn independently and uniformly from $[0, 90]$ CR — the 90 CR lease fee of §3's own running example is one realized draw from exactly this pair. Any mechanism satisfying Myerson–Satterthwaite's four desiderata on this bilateral slice is, by the theorem, not ex-post efficient; the canonical incentive-compatible, individually-rational, budget-balanced instance is the Chatterjee–Samuelson double auction [8], whose linear equilibrium on $[0, M]$ has Bob bid $\beta(v) = \frac{2}{3}v + \frac{M}{12}$ and Alice ask $\sigma(c) = \frac{2}{3}c + \frac{M}{4}$, trading iff $\beta(v) \geq \sigma(c) \iff v - c \geq M/4$. At $M = 90$ that threshold is 22.5 CR. Take Bob's value at 70 CR and Alice's cost at 60 CR: trade is efficient (a 10 CR surplus sits on the table) but $v - c = 10 < 22.5$, so the mechanism reports no trade — the efficient rental simply does not happen. Integrating over the whole square, the no-trade-but-efficient region $\{0 \leq v - c < M/4\}$ has probability $\dfrac{a(2M - a)}{2M^2}$ at $a = M/4$, which simplifies to $7/32 \approx 21.9\%$ independent of M: better than one trade in five that would leave both sides better off is refused by the very mechanism built to protect them from lying about v and c

[internal; algebra shown]. This is the concrete shape of Theorem 7.2's abstract impossibility, not a separate claim: raise the stakes (a bigger M) and the threshold and the missed trades grow with it, in the same fixed proportion.

The current design establishes a conservation goal and voluntary entry; it has not proved truthfulness. An AGV-style mechanism changes the incentive and participation notions and generally offers expected rather than ex-post balance. Comparing those alternatives requires a fully specified type and transfer model, which remains open.

Exercises 6.17–6.19, page 33.

Recall.

1. Without looking: state the composition law $\Delta(g \circ f) = ?$, and say which kind of ledger trace always has $\Delta = 0$.
2. What does a scalar cross-currency valuation need that native-unit conservation does not?
3. Name the four things Myerson–Satterthwaite says a bilateral trade mechanism cannot have all at once.

8 The Anchor Protocol proper: cross-harbor capability transfer

The *cross-harbor capability-transfer ceremony* is the market's identity-level hot path: how an authorization minted under Alice's harbor becomes usable, safely, inside Bob's. *The Anchor Protocol* analyzes the cryptographic substrate (a hop-bound authorization chain checked by protocol models plus a bounded Rust verification layer); *this* section is the ceremony that rides that substrate across a boundary.

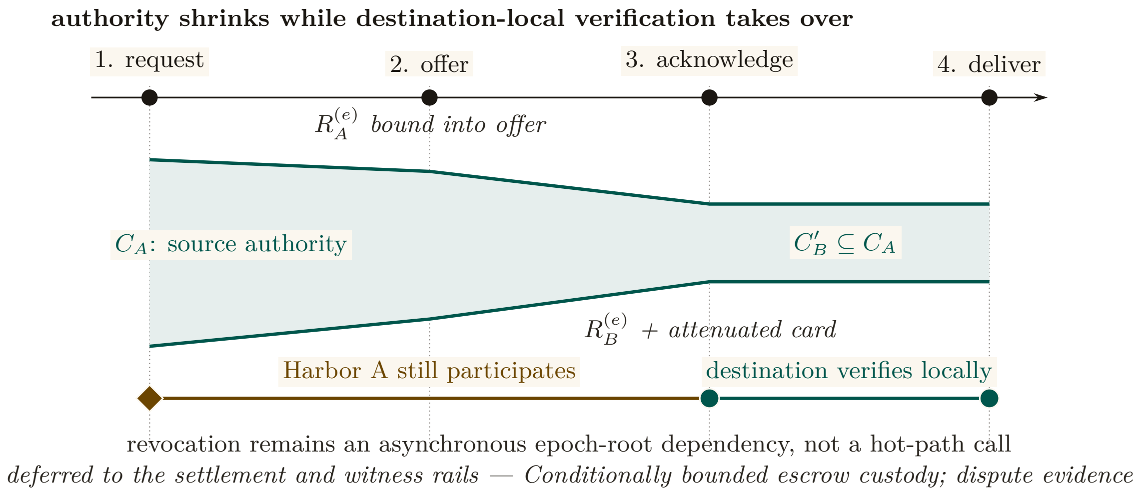

Figure 4: The transfer ceremony attenuates authority and then removes source reachback from the hot path. The green envelope narrows from C_A to $C'_B \subseteq C_A$ as the signed offer and acknowledgement bind source and destination epoch roots. After message 3, Harbor B can verify and deliver the attenuated card locally. Harbor A still matters asynchronously: a later epoch root can revoke the transfer once that root reaches B's audit surface. Transfer therefore neither copies full authority nor makes every use synchronous.

The four-message ceremony (Figure 4, written out as Protocol 8.1) has one decisive liveness property: after message (3), no further synchronous interaction with Harbor A is needed. The transferred card C'_B is verifiable under Harbor B's key alone, so federation does not turn every authorization into a synchronous inter-machine round trip. The epoch root $R_A^{(e)}$ bound into the envelope is the governing primitive: a revocation issued by A after epoch e invalidates C'_B as soon as the new root $R_A^{(e+1)}$ reaches the witness log that B audits.

Protocol 8.1 (Cross-harbor capability transfer)**. Parties:** Alice's agent (holds card C_A), Harbor A (Alice's daemon), Harbor B (Bob's daemon), Bob's agent (verifier). **Pre:** B audits a witness log to which A publishes epoch roots.

1. Alice's agent $\to$ A: xfer_req(C_A, B) — present C_A, name target harbor B, request a transferable form.
2. $A \to B$: xfer_offer$\langle C_A, \mathrm{att}, R_A^{(e)} \rangle$ — A signs an envelope binding C_A, an attenuation predicate att (which may only *narrow* authority), and its current epoch root $R_A^{(e)}$.
3. B $\to$ A: xfer_ack$\langle C'_B, R_B^{(e)} \rangle$ — B verifies A's key via the witness log, applies its *own* attenuation, and mints $C'_B = \mathrm{att}_B(\mathrm{att}_A(C_A))$, valid only at B.
4. $B \to$ Bob's agent: xfer_deliver$\langle C'_B \rangle$. Every later presentation verifies under B's key alone — no A-side lookup on the hot path.

Post (liveness): no synchronous A interaction after step (3). **Post (safety):** authority only ever narrowed; replay against a stale $R_A^{(e)}$ is rejected; a revocation by A kills C'_B once $R_A^{(e+1)}$ reaches B's witness log.

The ceremony, dramatized. Alice's specialist a_2 holds a card C_A that authorizes "read and write `src/payments/`, run the test suite." Bob rents a_2 for an afternoon. Alice's harbor offers an attenuated envelope: att_A strips write access to everything except the one module Bob hired for. Bob's harbor, not trusting Alice's word, attenuates *again*: att_B adds "no network egress, six-hour TTL." The minted C'_B is the intersection — read/write one module, run tests, no network, expiring at dusk — and it verifies under Bob's key, so a_2 never has to phone home to Alice mid-task. At 5 p.m. Bob's revocation fires locally; at the next epoch boundary Alice's harbor also sees, in the shared witness log, that C'_B was used exactly as attenuated and nowhere wider.

Key idea. Capability *attenuation* is the safety property here: a transferred card can only *narrow* authority — $C'_B = \mathrm{att}_B(\mathrm{att}_A(C_A))$ — never widen it. This is the cross-boundary form of the attenuation-monotonicity that the hop-bound authorization chain checks within one harbor (*The Anchor Protocol*). It is what makes "rent my agent for an afternoon" safe to say to a stranger.

Pitfall. Do not let the federation theorems borrow the substrate's "formally verified" halo. The hop-bound authorization chain has bounded symbolic and Rust evidence in *The Anchor Protocol*. The *federation* claims are design arguments plus a bounded TLA$^+$ extension; they are SPECIFIED, not deployed. Two different confidence levels, never blurred.

Exercises 6.20–6.22, page 33.

Recall.

1. Without looking: after which numbered message does Harbor A's continued participation stop being required?
2. State the attenuation-monotonicity property of C'_B in one sentence.
3. If A rotated its signing key after issuing, what must B check?

9 Federation: trading across machines you don't own

Federation is where the leaderless distributed canon actually bites. Crucially, the federation design *refuses consensus* rather than trying to route around Fischer–Lynch–Paterson — which is the correct reading of the impossibility the next box states.

Table 4 lists the four rails of the topology this section defends: independent harbors linked only by gossip and a shared witness log, with no global consensus step.

Table 4: Federation is four independent rails between two sovereign harbors, each keeping its own signing root; there is no shared root. The witness rail correlates and exposes equivocation but authorizes nothing; the capability and revocation rails carry attenuated authority and later invalidation in opposite directions; the settlement rail relates a bond posted at A to damage measured at B, and its custody bound is conditional on the settlement gate, not an invariant. No rail makes either harbor a root of authority for the other. [internal]

Rail	**What crosses**	**Direction**	**What it does not do**
witness	the signed epoch roots $R_A^{(e)}, R_B^{(e)}$ and any equivocation evidence	both ways	authorize work: correlation, not authorization
capability	an attenuated card, $C_A \mapsto C'_B \subseteq C_A$	A to B	widen scope or lifetime at any hop
revocation	the anti-entropy delta of revoked ids	B to A	promise a finite deadline under partition
settlement	a bond posted at A against damage measured at B	both ways	bound custody unconditionally: it holds only if the settlement gate holds

9.1 No consensus, by design

Key idea. A system that *needs* agreement on a global event order across operators it does not control is dead on arrival, by **Fischer–Lynch–Paterson** (1985) [16] (*no deterministic protocol achieves consensus in an asynchronous network with even one crash failure*). Federation declines to require agreement. It requires only (a) append-only per-harbor logs, (b) gossip-bounded propagation, and (c) *detectable* (not prevented) equivocation. This is a **Certificate Transparency** posture [18]: you do not prevent a misbehaving log from lying, you make lying detectable after the fact and expensive via a slashable bond.

The convergence and detection bounds smuggle in *partial synchrony* (**Dwork–Lynch–Stockmeyer** 1988 [17], the canonical escape from FLP): "gossip every Δ" *is* an eventual-synchrony assumption, and we state it as such rather than burying it.

9.2 Settlement and the cross-chain-deals lineage

Figure 5 places this construction against the atomic-swap and adversarial-commerce baselines it is compared to below.

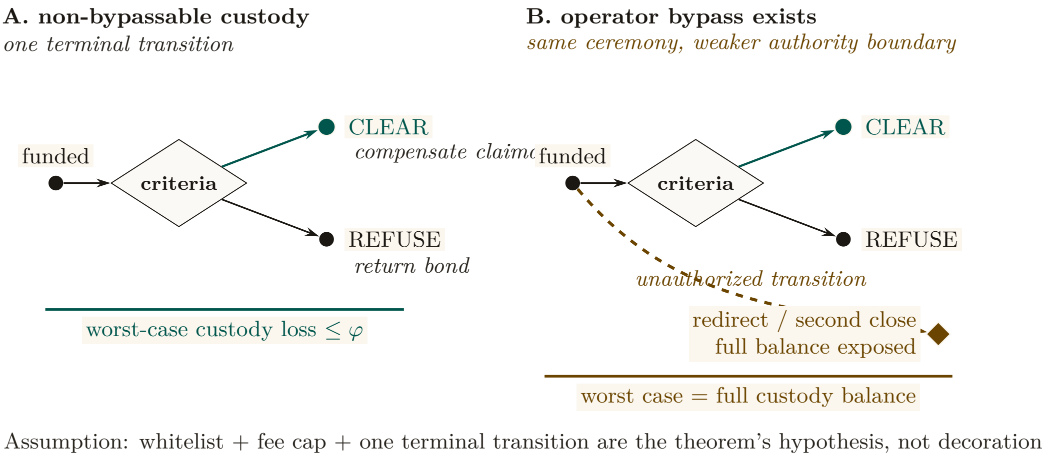

Figure 5: The custody bound is the absence of an extra transition. Panel A permits exactly one terminal move from funded custody to **Clear** or **Refuse**; with a recipient whitelist and fee cap, worst-case loss is bounded by the agreed fee φ. Panel B adds one operator-controlled bypass. The visible extra edge reaches redirect or a second close, exposing the full balance. Threshold signing is only one candidate implementation; the loss theorem holds only when the three authority restrictions are actually non-bypassable.

The escrow construction states its trust assumption outright: it does *not* claim trustless settlement. Its loss bound is conditional, and the condition is the whole content of the claim:

Design invariant 9.1 Bounded custody loss, conditional. *If* the custody ledger is non-bypassable — exposing only a recipient whitelist, a fee cap, and one atomic terminal transition, with no operator path around any of the three — then a counterparty's worst-case loss on a cross-harbor settlement is the pre-agreed fee φ, and the outcome space partitions into exactly **Clear** or **Refuse**. If that authority boundary *can* be bypassed, the worst-case loss is instead the full balance in custody. (SPECIFIED, conditional: no runtime conformance check establishes the hypothesis for a deployed system; 2-of-3 threshold signing is one candidate way to arrange it.)

This is a deliberate departure from the hash-timelock baseline (**Herlihy** 2018 [22], atomic cross-chain swaps), and the right comparison is the formal model of mutually-distrusting commerce without a shared chain: **Herlihy, Liskov & Shrira** (2022, *Cross-Chain Deals and Adversarial Commerce*) [23]. The harbor's bounded escrow is a "trusted third party / watchtower" point in that design space; we locate it there explicitly rather than reinvent it.

Remark. **Open problem: trustless cross-harbor settlement.** It is unknown here whether non-fungible, reputation-priced bonds admit a useful trustless settlement protocol under the paper's availability and privacy requirements. An HTLC comparison does not prove impossibility because the asset and adjudication models differ. (*open*.)

9.3 Revocation gossip and the equivocation race

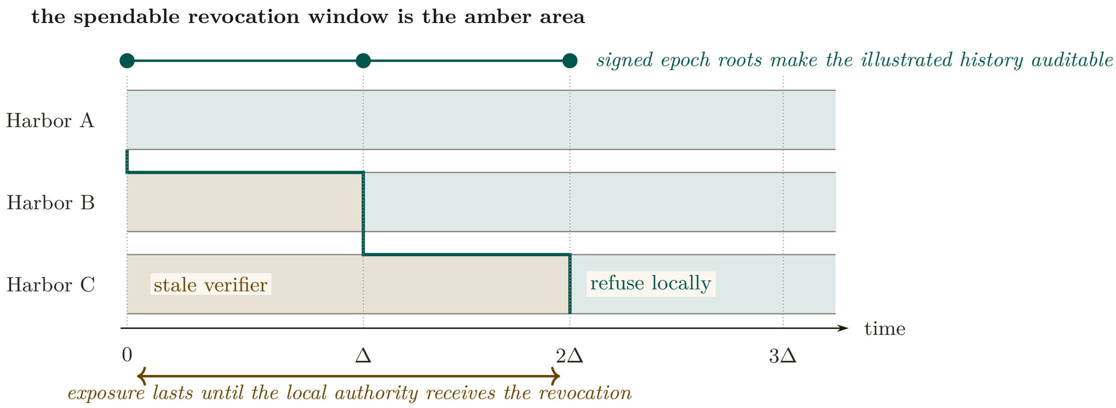

Figure 6: Revocation propagation is a moving staleness frontier. Alice refuses immediately; B remains stale until Δ and C until 2Δ in this illustrated execution. The amber area is precisely where a revoked capability can still be admitted by the named local authority. The teal staircase marks synchronization, not a global clock. Expected $\Theta(\log m)$ dissemination requires connected reliable rounds; a partition can extend the amber area without bound and therefore destroys any finite deadline.

Federated revocation propagates by anti-entropy gossip (**Demers et al.** 1987 [20]) with a probabilistic-membership filter (a cuckoo filter — a compact structure that answers "is this card revoked?" with no false negatives and a tunable false-positive rate). The federation therefore has expected $\Theta(\log m)$-round dissemination only under a connected reliable-round overlay with the stated randomized peer-selection model. This asymptotic expectation is not an exact constant or a partition-tolerant deadline.

Protocol 9.1 (Federated revocation gossip)**. State:** each harbor X keeps a revocation filter F_X and a current epoch root $R_X^{(e)}$ published to a shared witness log.

1. *Revoke.* An operator revokes card c at its home harbor A: $F_A \leftarrow F_A \cup \{c\}$; A advances to $R_A^{(e+1)}$ and publishes it.
2. *Gossip.* Every Δ, each harbor picks a peer and exchanges filter deltas (anti-entropy). A card present in either filter becomes present in both.
3. *Audit.* Any third party compares each harbor's published roots in the witness log; a harbor that publishes two contradictory roots for one epoch is *equivocating*. Verification is constant work once both signed roots are co-located; delivery to an auditor has the overlay's separate dissemination delay.

Post, conditionally: in the connected reliable-round model, every honest harbor learns the revocation in expected $\Theta(\Delta \log m)$ time. Under an unbounded partition there is no finite global-learning bound.

The revocation race, dramatized. Alice revokes a_2's card c at $t = 0$ (say she discovers it was misbehaving). Harbor B, which rented a_2, has not yet heard: for the interval $[0, \Delta]$, B's filter is stale and c is still spendable there. At $t = \Delta$ gossip reaches B, which adds c to F_B and refuses further use. Harbor C, two hops away, learns at $t = 2\Delta$. This is one illustrated execution; a different schedule or partition produces a different window. Conjecture 9.1 says the bond Alice posted on c must dominate whatever a_2 could spend at B inside that window, or a well-capitalized Bob could rationally race the gossip.

Pitfall. The threat is *adversarial*, not stochastic. An attacker who controls peer selection or partitions the gossip graph (an *eclipse attack*, **Heilman et al.** 2015 [21]) breaks the expected-time guarantee. The statement is conditional: a finite service-level bound needs eventual connectivity and a known post-stabilization delivery bound; epidemic analysis then supplies an expectation within that model, not an adversarial deadline.

Detection-after-the-fact deters only if the slashed bond exceeds the value extractable during the detection window. This is the analogue of the *cleanup lower bound* (the companion result that a cleanup bond must cover the worst-case cost of the damage it underwrites), and we state it as a target:

Conjecture 9.1 (Equivocation Lower Bound)**.** For a deployment that advertises a finite freshness window T, the witness bond must dominate the maximum value spendable against a revoked-but-not-yet-observed capability over T. If partitions are unbounded, no finite bond covers the worst case; the system must instead fail closed on stale roots or cap spend independently. (*open*; analogue of the cleanup lower bound.)

Figure 7 bands the window Conjecture 9.1 must price against the connectivity assumptions each band needs.

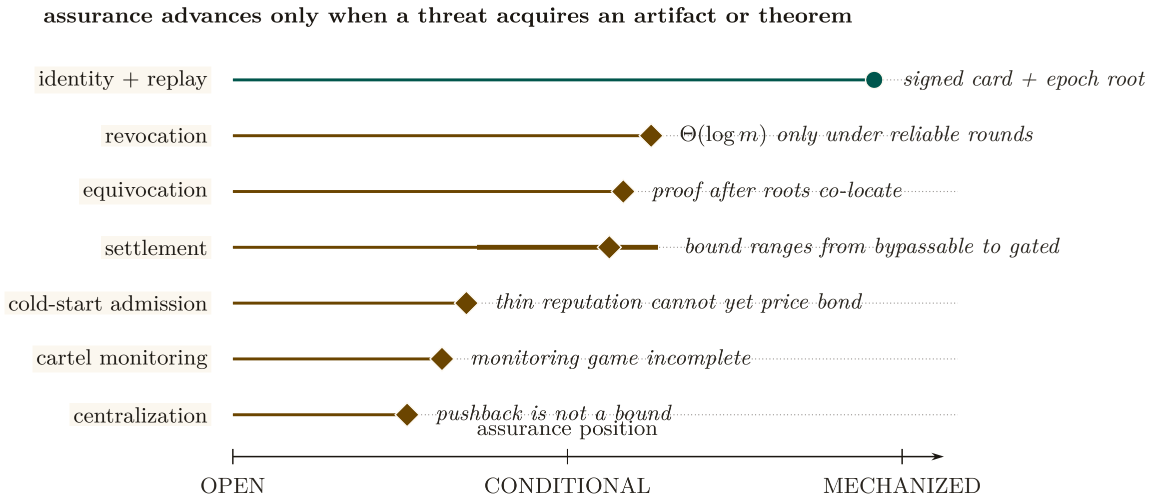

Figure 7: The assurance landscape is uneven and falsifiable. Position encodes how far each threat has moved from an acknowledged gap toward a mechanized artifact. Identity and replay have the strongest concrete evidence. Revocation, equivocation, and settlement are conditional on named connectivity or custody hypotheses; the thick settlement interval shows how widely its assurance moves if the gate is bypassable. Cold start, cartel monitoring, and centralization remain open. A row advances only when the missing artifact or theorem exists.

Exercises 6.23–6.25, page 34.

Recall.

1. Without looking: name the three things federation requires in place of consensus.
2. State the Bounded Custody Loss property's condition and its two possible outcomes.
3. Why is "gossip every Δ" itself a partial-synchrony assumption rather than a proven guarantee?

10 Reputation: monotone, but revocable

Reputation is the third side's existence condition. Two reconciliations matter here.

10.1 "Never ends" is not an anti-revocation property

A tempting design slogan holds that reputation "never ends" — no decay window an agent can outrun — while the same design also requires that a fraudulent settled outcome be retractable.

A property and its required violation cannot both be requirements.

Key idea. The reconciled statement: **reputation is monotone in *honest* witnessed outcomes, but a witnessed outcome is itself revocable via a propagating tombstone**. Dissemination has the same model and partition caveats as capability revocation. "Never decays" applies to honest outcomes only; it is *not* an anti-revocation property.

We also decline to assert no-decay as a virtue: bounded memory is a design parameter (**Liu & Skrzypacz** 2014, *Limited Records and Reputation Bubbles* [10], show unbounded records create reputation *bubbles*, not stability). The horizon is a tunable, and ∞ is rarely optimal.

10.2 Why Sagas-style compensation does not restore reputation

It is tempting to fix revocation with compensating transactions (**Garcia-Molina & Salem** 1987, *Sagas* [24]), as the bond ledger does. But the analogy fails structurally:

Theorem 10.1 Compensation does not erase reliance. A ledger can append a compensating transaction that restores a numerical balance. A reputation tombstone cannot undo decisions counterparties already made while relying on the invalid outcome. It can only change future evaluations after the tombstone is observed. Reputation repair therefore requires dissemination plus an explicit remedy for past reliance; it is not equivalent to ledger compensation.

This is why Sagas appears in the bond-settlement prior art but *not* in the reputation prior art.

10.3 The grading-oracle keystone

Every folk-theorem incentive-compatibility claim in the market bottoms out in "the daemon derives the grade." But the grade is produced by a capturable evaluator.

Theorem 10.2 The monitoring model must include the evaluator. An imperfect-public-monitoring equilibrium result requires a specified conditional distribution of public signals given modeled actions and strategies. If a strategic grader chooses the signal but is omitted from the game, that distribution is not fixed by the model and the claimed equilibrium result is unsupported. One must either constrain the signal mechanically or include the grader, its information, payoffs, and deviations as part of the game [9].

The two resolutions (both used). *Fix A — mechanically constrained evidence:* ground settlement in evidence the agent cannot directly author (a CI-issued test result, a protected merge event, or an independently evaluated policy check). This narrows the signal channel; it does not make the task metric universally strategy-proof. *Fix B — bond-and-slash the judges:* where machine-checking is impossible (aesthetics, ambiguous quality), the human/LLM judges get their *own* bond and their *own* reputation; a judge later contradicted by re-audit is slashed. The "rate-the-raters" recursion must then be shown to terminate at a machine-checkable base, not regress infinitely. De-biasing for LLM judges (order-swap, pairwise, family-exclude) follows **Zheng et al.** 2023 [14].

A captured judge, dramatized. Bob disputes a settlement: he claims Alice's a_2 delivered an "untasteful" refactor. Taste is not machine-checkable, so Fix B applies: Judy, a human grader, is drawn to arbitrate. Judy posts a 50 CR judge-bond and rules *for Bob*. But Judy and Bob are colluding — Judy is a captured judge, the exact failure Theorem 10.2 warns of. The market's defense is the re-audit lane: a sampled second panel later reviews Judy's ruling against the preserved evidence, finds it indefensible, and *contradicts* it. Judy's 50 CR bond is slashed and her judge-reputation takes a tombstone; Alice is made whole. The signal became exogenous again not because Judy was honest but because lying was bonded and the re-audit made it expensive — and the recursion stops at the second panel, whose own evidence is machine-checkable (the same diff, the same tests).

Exercises 6.26–6.28, page 34.

Recall.

1. Without looking: what CAN a reputation tombstone do, and what can it NOT undo?
2. State Theorem 10.2 in one sentence: what does an imperfect-public-monitoring result require that an omitted strategic grader leaves undefined?
3. Name the two resolutions (Fix A and Fix B) and the ground-truth base the recursion in Fix B must terminate at.

11 Hosted trust is the product

The 2025–26 agentic-payments landscape — **AP2** (Google, signed payment mandates), **ACP** (OpenAI/Stripe), and **x402** (Coinbase, stablecoin settlement over HTTP) — has commoditized the *settlement rail*. Wiring an agent to pay another agent is now table stakes. This is *good news* for the thesis, because it sharpens what the product actually is.

What remains scarce, and therefore defensible, is **hosted trust**: the verified ledger (conservation-checked, Merkle-anchored evidence), the *relay* (the harbor mesh carrying public-key attestation across machines), and the *reputation* that makes a counterparty you do not own legible and accountable.

- **Substrate (swappable):** the payment rail. Use x402/AP2/credits/ Stripe interchangeably.
- **Product (the moat):** attestation, federation membership, and reputation hosting.

The revenue model aligns incentives without perverse over-penalty: a transaction fee (2–5% of settlements), a small listing fee (collusion deterrent), and a small bond-spread on *forfeited* bonds (kept small so the platform never profits from over-slashing). The platform should earn most from *successful* trade, because successful trade is what hosted trust produces.
As §2 put it, before any of the machinery was on the page:

Thesis. You don't sell crypto — crypto is the substrate. Here is what *is* scarce: attestation + federation membership + reputation.

12 Cold start, and the auction question

A three-sided market has a *triple* cold-start problem: no operators shop an empty fleet shelf; no owners list agents with no renters; no licensors publish skills with no buyers. Two-sided-market theory prescribes: subsidize the side carrying the scarcest cross-side externality, then flip. Phase 1 seeds supply at zero/negative price with platform-posted tasks at above-market bounties (generating the first settled outcomes — the bootstrap reputation data); Phase 2 imports external work history for partial credit to break the new-agent freeze; Phase 3 flips to charging the demand side once a liquidity threshold (not a calendar date) is met. Figure 8 draws the schedule against the quantity that actually gates it, because "threshold, not date" is the kind of phrase that stays a slogan unless you can see what it is a threshold *on*.

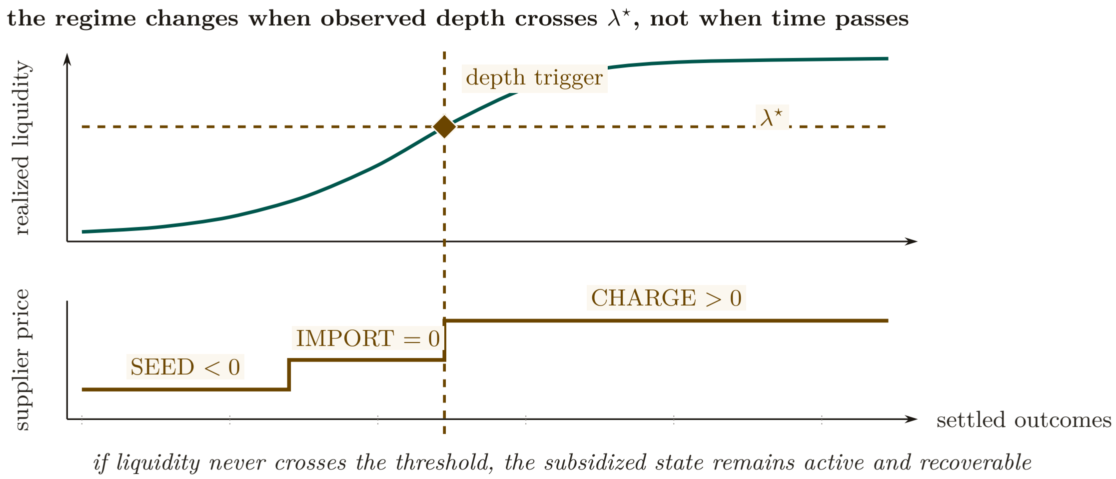

Figure 8: Cold start separates the measured market state from the policy schedule. The upper trace is realized liquidity over settled outcomes. The lower trace is the supplier-price regime applied to that state: seed supply below zero, credit imported work at zero, then charge only after liquidity crosses $\lambda^\star$. The shared vertical trigger makes the dependency explicit. Calendar time is absent, and a market that never crosses the threshold does not silently advance to the charging regime.

12.1 Static escrow vs. competitive underwriting

Thomas Youle's competitive-insurance proposal — insurer-agents bid to underwrite transactions, replacing a static bond parameter with a market-discovered price — is a possible *fourth* side. But it may collide with the cleanup lower bound:

Proposition 12.1 (Underwriting reframes the bond as reserve adequacy)**.** The cleanup lower bound requires bond $\geq$ worst-case cleanup cost c. A competitive insurance market prices to *expected* loss, which sits below the worst case for any below-tail agent. Either underwriting violates the cleanup lower bound (the commons can leak in the tail), or the bound must be reframed as the *insurer's* capital-adequacy constraint (a reserve-requirement framing with tail reinsurance), not a per-transaction bond. The *form* of the answer is determined; only the reserve formula is open.

12.2 Cartels and wash-trading

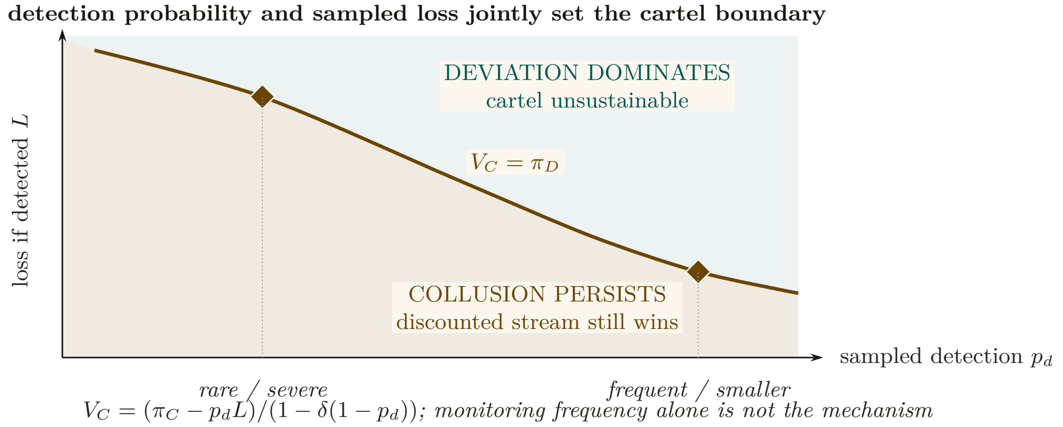

Figure 9: Cartel sustainability is a two-policy phase diagram. The boundary is the repeated-game equality $V_C = \pi_D$. Below it, the discounted collusive stream still dominates a one-shot deviation. Above it, deviation wins and the cartel is unstable. A rarer severe sampled loss and a more frequent smaller loss can lie on the same deterrence boundary. Detection frequency by itself is therefore not the mechanism; the product enters through the full discounted payoff expression.

Multi-homing plus portable reputation enables cross-harbor *cartels* — a rating-agency cartel is the closest real-world analogue: several independent principals colluding to hold a shared reputation signal artificially favorable rather than competing on it. And a colluding requester+worker can *wash-trade* fake settled outcomes to mint reputation cheaply. The defense form — cost-per-settled-outcome must exceed reputation's marginal value — is circular as stated, because that marginal value is endogenous to the same market. The fix is a structural break: make settled outcomes between a counterparty *pair* exhibit sharply diminishing reputation returns (a concavity that caps the wash-trade payoff regardless of price), plus sampled adversarial re-audit of high-velocity pairs.

Whether a cartel of colluding operators can actually hold its rating floor is a *repeated* game, and that is what Figure 9draws. Each round a member either *colludes* (holds the agreed floor q_{floor}, collecting the cartel payoff π_C) or *defects* (undercuts the floor by ε for a one-time gain π_D, after which the cartel unravels and everyone drops to the competitive payoff $\pi_N \approx 0$). A **grim-trigger** strategy is the crudest way to punish that: *cooperate until the first observed defection, then defect forever.* It sustains the cartel only while members weight the future heavily enough — that weight is the **discount factor** $\delta \in (0, 1)$, the value a member places on next round's payoff relative to this one's, so a member with δ near 1 is patient and one with δ near 0 will take π_D today and let the cartel die. Sustaining collusion means the discounted stream of colluding payoffs beats the one-shot deviation, which is exactly the inequality the figure's box states, and δ clearing its threshold is exactly what the defenses above are trying to prevent.

Exercises 6.29–6.31, page 34.

Recall.

1. Without looking: which side does Phase 1 subsidize, and by what rule?
2. State Proposition 12.1's two horns for what happens once a competitive insurance market prices to expected rather than worst-case loss.
3. What structural change — not a price threshold — is needed to cap the wash-trading payoff regardless of price?

13 A gluing analogy for the open federation problem

Three open problems—cross-harbor settlement, equivocation under partitions, and cross-currency valuation—share a local-to-global shape. That resemblance motivates an analogy; it does not yet define a sheaf or a cohomology theory.

Conjecture 13.1 (Formalization target, not established result)**.** A future treatment may define a site of overlapping administrative domains and a presheaf of signed ledger views, then ask when compatible local sections admit a unique global section. This paper has not specified the site, restriction maps, coefficient object, or cocycles; therefore it makes no claim that a particular H^1 is defined or non-zero. The immediate engineering obligations remain the concrete ones: matched custody transitions, freshness policy under partitions, and time-indexed valuation.

Fong and Spivak's compositionality program [26] suggests vocabulary for a future formalization. Until the site and coefficient data are defined, the paper uses "gluing" only to organize engineering questions.

Exercises 6.32–6.34, page 34.

Recall.

1. Without looking: name three of the four things Conjecture 13.1 says this paper has not specified about the proposed gluing model.
2. Why are two incompatible signed roots evidence of equivocation but not, by themselves, a cohomology class?
3. What are the concrete engineering obligations that remain regardless of whether the formalization is ever completed?

14 Gaps the design must still close

Naming what is unspecified is part of the specification. These are not threats already defended; they are holes.

1. **Multi-dimensional reputation aggregation.** "Accuracy/aesthetics/ efficiency judged by separate experts" has no specified combination rule. You cannot have "separate axes" and "one buyer-readable score" without specifying an aggregation function and its trade-offs. (*proposed*.)
2. **Float-plan amendment under scope drift.** The four states assume the acceptance criteria were right. Mid-flight re-specification needs a re-sign protocol with escrow top-up/refund delta, preserving conservation.
3. **Cross-boundary key rotation.** How a rotated signing key invalidates or re-validates cross-harbor delegations already held by B is unspecified.
4. **Skill versioning.** Reputation attaches to a content hash; v2 starts cold. The lemons problem reappears at every version bump.
5. **Operator exit / harbor death.** Escrow orphaning, in-flight float plans, and reputation tombstoning on ungraceful exit are unspecified — the market-level analogue of the kernel's crash-recovery problem.
6. **Incentive-compatible arbitration capacity.** The human oracle is scarce and expensive; at scale arbitration is itself a priced, bondable, slashable service that must converge to honest rulings, not rubber-stamping.

These six are not equally dangerous, and an implementer triaging them should not have to infer the stakes from the prose. Each one threatens a specific critical promise made earlier in this paper:

Table 5: Which promise each gap threatens. The two gaps that reach the *conservation* invariant are the urgent ones, because conservation is the only claim in this paper graded **implemented** — a hole there falsifies something that is currently true, rather than deferring something that is not yet true.

Gap	Threatens	How it bites
Operator exit / harbor death	conservation	orphaned escrow rows have no settlement path, so wallet + escrow + commons stops summing to supply
Float-plan amendment under scope drift	conservation	a mid-flight re-spec moves value with no matching debit unless the top-up/refund delta is escrowed
Cross-boundary key rotation	cross-boundary safety	a delegation already held by B may verify against a key A has retired
Skill versioning	Sybil-resistance	a v2 content hash is a fresh identity, so the lemons problem and the whitewash both reappear at every version bump
Multi-dimensional reputation aggregation	incentive compatibility	an unspecified aggregation rule is an unspecified target, and any fixed scalar collapse re-creates a Goodhart objective
Incentive-compatible arbitration capacity	incentive compatibility	Fix B (§10) prices one judge; it does not price a judge pool under load, where rubber-stamping is the cheap strategy

Review of the key ideas

What this chapter leaves coupled, in the order the rest of the book will lean on it:

1. **A market, not a payment rail.** The harbor prices work between three sides, an operator-for-hire, a buyer and an insurer, and the escrow row is the one object all three sides read.
2. **The succession rule has a price.** Letting a worker be replaced mid-task costs $D^* = \eta K/(1-\eta K)$ in expectation; below that, the buyer is better off waiting.
3. **The float plan is the built floor.** A worker becomes *running* only at the committed transition where the wallet is debited and the escrow row exists; a signed plan alone admits nothing, and abort leaves all three traces unchanged.
4. **Conservation is an invariant, not a hope.** In native units $\Delta W + \Delta E + \Delta C = 0$ exactly, and no spawn happens without a bond; once amounts are converted, fee, slippage and risk are priced terms and the total is no longer zero by construction.
5. **Deontic conflict is decidable inside the fragment and NP-complete one step outside it.** The in-fragment check is polynomial and produces a witness; that is what the daemon runs.
6. **The keystone is missing and named.** Cross-operator attestation is what the market rests on and does not yet have; local identity is built.
7. **Single-unit conservation composes**; bilateral trade across a boundary is conditional, and the condition is stated.
8. **Reputation is monotone but revocable**: compensation does not erase reliance, and the monitoring model must include the evaluator.

9. **Hosted trust is the product**: cold start is solved by a supplier price at λ^*, not by an auction the market cannot yet clear; the cartel game says which regime detection and loss put an insurer in.

15 Exercises

§2 The thesis: a market, not a payment rail

6.1 Check ⋆ Give a one-sentence reason the market cannot ship before reputation does, using only the word "clone." *(solution on page 40)*

6.2 Check ⋆ Side 1 (operator-for-hire) does *not* require durable identity in the same way sides 2 and 3 do. Why? (Hint: who is the counterparty, and who bears the consequence?) *(solution on page 40)*

6.3 Open ⋆⋆⋆ Construct a market design in which sides 2 and 3 exist *without* a non-forgeable identity. Argue informally why every such design collapses to a Sybil attack (forward-reference §6). *(solution on page 41)*

§3 What a "side" is, and why there are three

6.4 Check ⋆ State the operational test for "how many sides" in one sentence, then apply it to a ride-share platform: how many sides, and why? *(solution on page 41)*

6.5 Trace ⋆⋆ Side 3 (skill licensing) keys reputation on a *content hash*, not a person. What breaks when a licensor ships v2 of a skill? (Hint: §14, skill versioning.) *(solution on page 41)*

6.6 Open ⋆⋆⋆ A buyer reads a three-axis reputation vector (accuracy, aesthetics, efficiency). Specify a disclosure rule that lets the buyer weight the axes *without* re-introducing a single Goodhart target. Argue why every fixed scalar collapse re-creates the bandit framing the design rejects. *(solution on page 41)*

§4 The float plan: the built floor

6.7 Trace ⋆⋆ Trace the flow of funds for a **Partial** settlement where the worker completed 40% of the evidence chain. Confirm Design Invariant 4.1 holds across the split (the dramatization above is one such trace; redo it for a bond of 250 CR and 70% completion). *(solution on page 41)*

6.8 Check ⋆ Why does closure bind to an oracle rather than a self-reported note? Name the attack a free-text "Result:" note enables. *(solution on page 41)*

6.9 Open ⋆⋆⋆ The four states assume the acceptance criteria were *right*. Design a *float-plan amendment* protocol for the common case where the criteria themselves turn out wrong mid-flight (re-sign by both parties, escrow top-up/refund delta), preserving Design Invariant 4.1. (Gap, §14.) *(solution on page 41)*

§5 Three sides on one escrow

6.10 Check ⋆ Map each of Grossman–Hart's two parties (residual-control-right holder vs. reputational-consequence bearer) onto the rental side's roles. Which one is the Arbiter jail? *(solution on page 41)*

6.11 Trace ⋆⋆ Show that adding a lease line item and a metered-license line item to a float plan cannot violate Design Invariant 4.1, no matter the amounts. *(solution on page 42)*

6.12 Open ⋆⋆⋆ A skill's portfolio proof for v1 says nothing about v2. Specify a *version-binding* for the Merkle portfolio proof so a new version starts with a discounted inherited prior — the newcomer policy, but for code. *(solution on page 42)*

6.13 Trace ⋆⋆ The deontic layer named above — what an agent *may* or *must* do, as against the jail's *capability* — needs a decision procedure once two accepted norms might disagree. Take the conflict witness worked by hand above (fact `agent_deployed`; Horn rule to `prod_access`; an obligation and a forbiddance on `write_prod` overlapping on $[5, 10]$), and add one *disjunctive* obligation $O(\ell_1 \vee \ell_2)$ over two fresh literals unrelated to the existing rules. Try the three nonempty discharge selections (ℓ_1 alone, ℓ_2 alone, both) by hand. Does any selection make the `write_prod` conflict go away? What happens on the selection that asserts both ℓ_1 and ℓ_2, and why does Theorem B4.2's reduction from 3-SAT depend on exactly that happening? *(solution on page 42)*

§6 The keystone the market rests on — and does not yet have

6.14 Check ⋆ In one sentence, state why a reputation system layered on forgeable pseudonyms is *no mechanism*, not merely a weak one (cite Douceur). *(solution on page 42)*

6.15 Check ⋆ Local non-forgeable identity has a non-goal: "no defense against a hostile operator." Name the exact word in the market's definition ("mutually-distrusting," "boundary neither controls") that this non-goal contradicts. *(solution on page 42)*

6.16 Open ⋆⋆⋆ Specify a cryptographic binding tying N agents to one principal such that spawning fresh agents does not reset the bond floor, and that survives into the cross-operator world (where the single-operator "trusted credential" shortcut is unavailable). This is the hard half of cross-operator attestation. *(solution on page 43)*

§7 Conservation under composition

6.17 Check ⋆ State Myerson–Satterthwaite in one sentence and list the assumptions that must be checked before applying it to one harbor trade. *(solution on page 43)*

6.18 Trace ⋆⋆ Give a two-currency example in which native-unit conservation holds but the marked-to-market scalar changes because v_t moves. *(solution on page 43)*

6.19 Open ⋆⋆⋆ Specify an oracle-latency and price-move model that yields a valid φ exposure bound for Property 7.1. *(solution on page 43)*

§8 The Anchor Protocol proper: cross-harbor capability transfer

6.20 Check ⋆ Why is the ceremony four messages and not two? Identify what message (3) buys (Hint: who must counter-sign, and under whose key does C'_B later verify?). *(solution on page 43)*

6.21 Trace ⋆⋆ Trace what happens to an in-flight C'_B when Alice rotates her signing key between messages (3) and a later presentation at B. (Gap: cross-boundary key rotation, §14.) *(solution on page 43)*

6.22 Open ⋆⋆⋆ Specify replay protection across the boundary: is the *anchor id* a nonce? Does B reject a re-presented message (3)? State the freshness invariant. *(solution on page 44)*

§9 Federation: trading across machines you don't own

6.23 Check ⋆ State the connectivity, peer-selection, and delivery assumptions needed for expected $\Theta(\Delta \log m)$ dissemination. Explain why none yields a deadline during an unbounded partition. *(solution on page 44)*

6.24 Trace ⋆⋆ Trace a revoked capability through Figure 6: at which harbor, and at what time, can it still be spent? That window is what Conjecture 9.1 must price. *(solution on page 44)*

6.25 Open ⋆⋆⋆ State and (attempt to) prove the Equivocation Lower Bound for a three-harbor mesh under a fixed gossip period Δ and a single equivocating witness. *(solution on page 44)*

§10 Reputation: monotone, but revocable

6.26 Check ⋆ State the signal-model assumption Theorem 10.2 needs and explain why an omitted strategic grader leaves it undefined. *(solution on page 44)*

6.27 Check ⋆ Classify three settlement criteria as machine-checkable (Fix A) or judge- dependent (Fix B): a unit test passing; a merged SHA; "the refactor is tasteful." *(solution on page 44)*

6.28 Open ⋆⋆⋆ Prove the rate-the-raters recursion terminates: define a well- founded order on "levels of judge," grounded at machine-checkable oracles, and show no infinite ascending chain of "judges judging judges" is needed. *(solution on page 44)*

§12 Cold start, and the auction question

6.29 Check ⋆ Which side should Phase 1 subsidize, and why? Apply the "scarcest cross-side externality" rule. *(solution on page 45)*

6.30 Trace ⋆⋆ Use Figure 9to identify the grim-trigger condition under which the cartel is self-sustaining. What raises the critical discount factor δ enough to break it? *(solution on page 45)*

6.31 Open ⋆⋆⋆ "Price of anarchy when reputation is for sale" is the wrong tool: once reputation is tradeable, you have changed the game, so PoA (a ratio for a *fixed* game) does not apply. Reframe it as a *signal-existence* question (Spence-signaling with a resale market): does *any* informative separating equilibrium survive reputation resale? *(solution on page 45)*

§13 A gluing analogy for the open federation problem

6.32 Check ⋆ Name the site, cover, restriction maps, and coefficient object a genuine sheaf model of federated ledger views would require. *(solution on page 45)*

6.33 Check ⋆ Explain why two incompatible signed roots are evidence of equivocation but are not, without further definitions, a cohomology class. *(solution on page 45)*

6.34 Open ⋆⋆⋆ Build a small formal gluing model for a bounded gossip topology and state exactly what consistency result it proves. *(solution on page 45)*

16 Related work

The design borrows from seven literatures, and its only originality is in how it *joins* them; this section consolidates the threads cited piecemeal above.

Multi-sided platform markets. The frame is Rochet & Tirole [1, 2] and Armstrong [3]: a platform is multi-sided when the *allocation* of price across user groups, not just its level, moves volume. We extend the standard two-sided picture to three sides by counting *incentive constraints*, and we read the contemporary AI-labor-market model of Chiu, Zhang & van der Schaar [11] as confirmation that "reputation system as a third side" is the natural structure.

Mechanism design and its impossibilities. Settlement is a direct mechanism in the sense of Myerson [6]; the binding constraint is Myerson–Satterthwaite [7], which forces the market to surrender efficiency to keep budget balance, incentive compatibility, and individual rationality; the worked no-trade region in §7.1 instantiates the impossibility on the linear double auction of Chatterjee & Samuelson [8]. The incentive-compatibility arguments are imperfect-monitoring folk theorems (Green–Porter; Abreu–Pearce–Stacchetti [9]), whose exogenous public signal we cannot take for granted — hence the grading-oracle problem. Grossman–Hart [4] supplies the residual-control-right argument for the rental side, and Akerlof [5] the lemons argument for licensing.

Reputation systems. Empirically, online reputation has measurable value (Resnick et al. [12]; Tadelis [13]); theoretically, unbounded records create reputation *bubbles* rather than stability (Liu & Skrzypacz [10]), which is why our horizon is a tunable. The Sybil attack [15] is the reason reputation must key on a principal, not a spawn. LLM-as-judge de-biasing follows Zheng et al. [14].

Distributed systems and cross-chain commerce. Federation refuses consensus because of FLP [16], recovers a weaker guarantee under partial synchrony [17], propagates by epidemic gossip [20], and takes a Certificate-Transparency detect-don't-prevent posture [18, 19], mindful of eclipse attacks [21]. The settlement design is located explicitly against atomic cross-chain swaps [22] and the adversarial-commerce model of Herlihy, Liskov & Shrira [23]; reputation revocation is contrasted with Sagas-style compensation [24].

Applied category theory. The trace-composition notation and proposed gluing formalization draw vocabulary from ologs [25] and Fong & Spivak [26]; no cohomology claim is made. The commons-governance reading draws on Ostrom [27], and the auction machinery on standard algorithmic game theory [28].

Agentic-commerce protocols. While this design was being written, the industry shipped the transaction layer of an agent economy. The Universal Commerce Protocol [29] (Google/Shopify, 2026) standardizes agent-mediated retail — discovery via a `/.well-known` capability profile, a checkout state machine, server-selects version negotiation — and its rival, the Agentic Commerce Protocol [31] (OpenAI/Stripe), the platform-mediated equivalent. Both are deliberate about what they exclude: neither settles payments (delegated to payment service providers) and neither decides which agents deserve trust. Authorization proof is delegated to the Agent Payments Protocol [30], whose mandates — W3C Verifiable Credentials in SD-JWT form, chained principal → platform → merchant — are the deployed sibling of this paper's float-plan countersign, and whose credential formats the reference implementation now adopts at the harbor boundary (ADR-0094) so that the keystone identity artifacts of §6 verify with off-the-shelf tooling. The gap those protocols leave open is this paper's subject: published security analyses of UCP [32] note that it regulates *how* agents transact while offering nothing about *which* agents to trust — no collateral, no settlement oracle, no reputation, no priced deterrent. The harbor economy is a mechanism for exactly that layer; the retail protocols are evidence the layer below it is arriving on schedule, and hosted trust (§11) is what they will need bolted on.

Interactive proofs and adversarial debate. The adversarial test market (§17) is a seventh thread, cited piecemeal there rather than above because it grounds a single additional mechanism rather than the market's core structure. It is delegated computation in the sense of Goldwasser, Kalai & Rothblum [33] — a resource-bounded verifier need not redo a prover's work — crossed with multi-agent debate [34], where two adversarial reasoners are more legible to a judge than one cooperative one. Becker's deterrence condition [35] supplies the enforcement half: a test-writer's expected forfeiture must match its gain from misreporting.

Table 6: How the harbor economy sits relative to the nearest prior art on each axis.

Axis	**Nearest prior art**	**What the harbor does differently**
Market structure	Two-sided platforms [1]	Three incentive constraints (labor / capital / IP) on one escrow object
Settlement across distrust	Atomic swaps [22]; cross-chain deals [23]	Bounded (not trustless) escrow for *non-fungible, reputation-priced* bonds
Revocation	PKI revocation lists; Certificate Transparency [18]	Gossip-converged filters with a priced, named attacker window
Reputation revocation	Sagas compensation [24]	Tombstone propagation — no conserved quantity to compensate
Grading / IC signal	Folk theorems [9]	Strategy-proof machine-checkable oracle *or* bonded-and-slashed judges
Agent transaction rails	UCP [29]; ACP [31]; AP2 mandates [30]	They standardize the transaction and name trust out of scope; the harbor prices and enforces the trust layer they delegate away
Purchased assurance	Delegated computation [33]; AI-safety debate [34]	Prices assurance as k independently bonded test-writers under a Becker-style re-audit rate, against a mutation-testing oracle

17 One further market: adversarial testing and purchased assurance

Beyond labor, capital, and IP licensing, the harbor supports one further internal market, and it is worth stating because it is the one place where *assurance itself* becomes a priced line item rather than a hope. In an **adversarial test market**, independent test-writing agents compete to find flaws in a proposed diff *before* it settles. The frame is an interactive proof system: delegated computation [33] (a prover convinces a resource-bounded verifier without the verifier redoing the work) crossed with multi-agent debate [34] (two adversarial reasoners are more legible to a judge than one cooperative one).

The adversarial test market, dramatized. Bob's float plan ships a diff that one of Alice's specialists wrote. Rather than trust Alice's own test run — a signal whose author has every reason to flatter it — Bob posts a 30 CR bounty per confirmed flaw and opens the diff to $k = 3$ independent test-writing agents, each paid only for a defect it actually surfaces. If each catches a given real defect independently with probability $d = 0.6$, the chance a planted flaw survives all three is $(1 - 0.6)^3 = 0.064$. Bob has bought his defect risk down from 1 to about 6% for at most $3 \times 30 = 90$ CR, which is a number he can hold up against what the bug would have cost him in production. Raise k to 5 and the survival probability falls to $(0.4)^5 = 0.010$; that is the whole shape of the mechanism — assurance is bought in units of k, and each unit costs one bounty plus the carry on one bond.

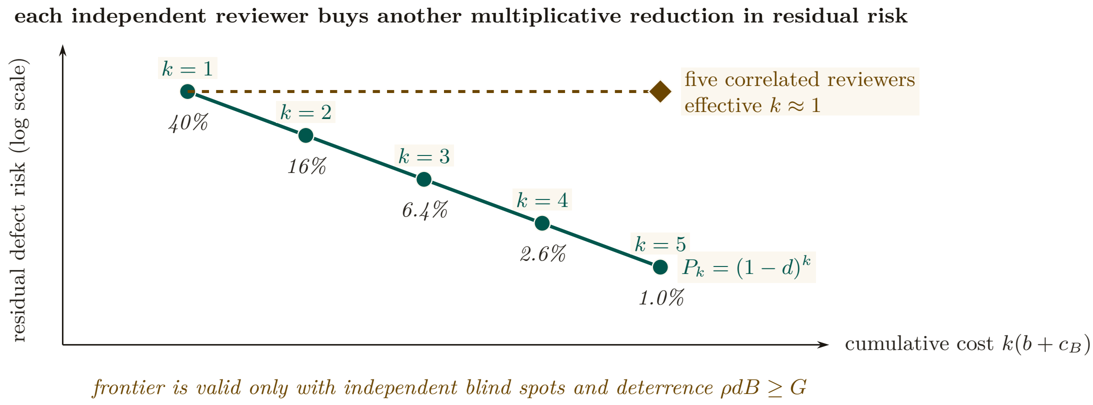

Figure 10: Purchased assurance is one risk–cost trajectory. Each teal point adds one independently bonded reviewer: cumulative cost rises linearly while residual defect risk falls geometrically. At $d = .6$, three reviewers leave 6.4% residual risk and five leave about 1%. The dashed amber counterfactual charges for five correlated reviewers but delivers roughly one reviewer's protection. Independence and Becker deterrence $\rho dB \geq G$ are therefore conditions of the frontier, not footnotes to headcount.

A **Becker-enforced panel** is a set of k independent, conflict-free test-writing agents, each detecting a given flaw with probability $d \in (0, 1)$, each posting a slashable bond B to enter the market, and each facing a platform re-audit rate ρ against a one-shot misreporting gain G. The panel size k and the detection probability d fix what the panel *catches*; the triple (G, B, ρ) separately fixes whether a test-writer is honest at all. The two questions are independent, so the paper states them as two results rather than one bundled claim.

Theorem 17.1 Purchased Assurance Bound. Let an author-agent submit a float-plan implementation with a proposed diff, and let a Becker-enforced panel of size k review it. Then the probability that a critical flaw survives undetected across all k reviewers is bounded by

$$\Pr(\text{defect undetected}) \;\leq\; (1-d)^k. \tag{1}$$

The operator's assurance level $1-(1-d)^k$ is then a purchasable commodity priced at $k \cdot (\text{bounty} + \text{bond carry})$, which is what makes "hosted trust as a service" a quantifiable line item rather than a slogan.

Theorem 17.2 Becker deterrence for the panel. Truthful effort is a dominant strategy for a Becker-enforced panel's test-writers exactly when expected punishment at least matches the one-shot gain from cheating — the **Becker deterrence condition** [35]:

$$\rho \, d \, B \;\geq\; G, \tag{2}$$

equivalently, the platform must commit to a re-audit rate at or above the critical rate $\rho^\star = G/(dB)$. (SPECIFIED; this half is imported, not proved here — see the paragraph below.)

The deterrence condition in Theorem 17.2 is not proved here; it is imported. The companion technical paper *Reputation is Amortized Verification* [36] derives $\rho^\star = G/(dB)$ as the stage-game threshold of a bonded inspection game — the same game this paper plays in §10 when it bonds and slashes a judge — and works it at the program's running parameters: $G = 10$, $d = 0.8$, $B = 50$ give $\rho^\star = 10/(0.8 \cdot 50) = 0.25$. Audit one report in four and the cheat payoff is exactly

zero. Double the bond to $B = 100$ and $\rho^\star$ halves to 0.125: capital and vigilance are substitutes along the frontier $\rho dB = G$.
In the notation of the audit tower of *From Spawn to Person*, this is the single-level case. There the bribe floor is written $\beta = \rho dB$, the gain at level k is G_k, and level $k{+}1$ audits level k across C sealed cliques, so the condition here reads $\beta \geq G_0$ and the contraction $G_{k+1} = (1 - \rho d)\, G_k$ is what makes a finite stack of audited auditors converge. The letter k is overloaded between the two chapters: here it counts the panel, there it indexes the level.

Pitfall. Quote the threshold without naming your settlement convention and you will be reproducibly wrong by about 17%. Equation (2) is the *keep-gain* convention: a caught cheat forfeits B but keeps G. If the escrow also claws the gain back, the forfeiture is $G + B$ and the critical rate drops to $\rho_c^\star = G/\big(d(G + B)\big) = 0.2083$ at the same parameters. A platform whose rail does claw back but which budgets audits at $G/(dB)$ is over-auditing; one that does not claw back but budgets at $G/(d(G{+}B))$ is under-deterring. The convention is a property of the slashing contract, not a modeling taste [36].

Pitfall. The $(1 - d)^k$ bound is exactly as good as the word *independent* in its hypothesis, and independence is the assumption a real test market breaks first. Three test-writers that read each other's public findings, or that are three instances of the same model family with the same blind spot, do not give k independent draws — they give something closer to one draw and two echoes. Purchased assurance therefore has to buy *diversity* (distinct principals, distinct model families, sealed submissions) and not merely count k; otherwise the priced assurance level is an overstatement of the delivered one. The same requirement appears in [36] as the sealed multi-clique sampling hypothesis, where it is likewise required rather than hygienic.

The ground-truth oracle for grading the test-writers themselves is **mutation-testing kill rates** against held-out syntactic mutation operators, which stops agents from overfitting tests to a suite they can already see. This is a Fix A oracle in the sense of §10: machine-checkable, and not authored by the party being graded.

(*proposed*. The mutation-testing oracle and the bounty-per-killed-mutant contract are specified here for the first time in this series; neither ships, and neither appears in the reference implementation's ledger module. The deterrence inequality it leans on is SPECIFIED, proved in [36]. Recorded as such in Table 7.)

Recall.

1. Without looking: state the Purchased Assurance Bound in one line, and name the one word in its hypothesis a real test market breaks first.
2. What is the difference between the Becker deterrence condition's "keep-gain" and "claw-back" conventions, and why does it matter which one you quote?
3. What is the ground-truth oracle for grading the test-writers themselves, and why does it have to be Fix A rather than Fix B?

18 Conclusion

The harbor-master makes a swarm legible and accountable for one operator (the wedge of *The Legible Swarm*). When operators sail out to trade, the *same* ledger that kept one operator's swarm honest becomes the market that lets fleets who don't trust each other work together: three sides, one bond, hosted trust. We have not papered over the cost of that claim. The keystone identity primitive — local non-forgeable identity — covers a single-operator threat model and must be extended to cross-operator attestation before the boundary is formally one "neither controls." Conservation composes exactly within each unit of account; cross-currency totals require time-indexed valuation and an explicit exposure model. Reputation is monotone in honest outcomes and revocable by tombstone, not by Sagas. Every folk-theorem result rests on a grading process included in the monitoring model. Myerson–Satterthwaite constrains bilateral slices only when its assumptions hold; the paper does not yet prove which corner the full market sacrifices. The bones are good and the floor is built; the spine the market trades on has partial local identity and outcome substrates but no complete reputation or cross-operator binding — and this paper says so in the same words throughout, because a market built on an overstated keystone is a market built on sand.

A Implementation & status

The body of this paper is deliberately implementation-agnostic. This appendix records, for the reader who wants to check the claims against running code, exactly which mechanisms are shipped in the reference implementation, which are specified with a proof or a written protocol, and which are still proposed. The single-line summary: an **implemented** conserving bond ledger, an **implemented** continuity organ (episodic memory), a PARTIAL checkpoint, and a richly SPECIFIED-and-model-verified protocol stack — but the spine the market trades on (cross-operator identity → outcome ledger → reputation) is PARTIAL-to-*proposed*. Today the harbor economy is a two-sided market with a proof-backed roadmap to three.

Table 7: Per-claim maturity, with concrete grounding in the reference implementation. "Module" names refer to source files in the implementation; "model" means a mechanized proof artifact; "property test" means a randomized test exercising the invariant.

Claim / mechanism	State	Grounding
Bond ledger: escrow, slash, commons pool	**implemented**	bond-ledger module (`lib/bonds.ts`)
Conservation wallet + escrow + commons = supply	**implemented**	10k-trace property test
No-spawn-without-bond	PARTIAL	escrow-first; runtime gate is a roadmap item
Episodic memory (continuity organ 1)	**implemented**	episodic-memory module
Resurrection / checkpoint (organ 2)	PARTIAL	passes notes, not state
Outcome ledger (organ 3 — reputation keys here)	PARTIAL	commitments, daemon-derived deadlines, oracle closure, API/CLI, and overdue monitoring ship; neutral grading and reputation binding do not ship
Local non-forgeable identity (intra-harbor)	PARTIAL	`actor-souls`, lookup credentials, and a bounded newcomer pool ship; full write gating does not ship
Cross-operator attestation	SPECIFIED/*proposed*	no spec for the critical part
Float plan + signed escrow ceremony	SPECIFIED	Protocol 4.1
Cross-harbor capability transfer (attenuation)	SPECIFIED	verified *as model*; Protocol 8.1
Reputation estimator (Elo / Bradley–Terry / TrueSkill)	SPECIFIED/*proposed*	no estimator shipped
Multi-dimensional reputation, neutral judges	*proposed*	no axis-aggregation spec
Three-sided market (labor / rental / licensing)	SPECIFIED	two-sided in reality
Skill licensing (metered + clawback + Merkle proof)	SPECIFIED	cross-harbor verifiability undeployed
Adversarial test market (Purchased Assurance Bound, Thm. 17.1)	*proposed*	no mutation-testing oracle and no bounty-per-killed-mutant contract shipped; the deterrence threshold it cites is proved in [36]

(continued)

Claim / mechanism	**State**	**Grounding**
Federation: witness log, capability transfer	SPECIFIED	*The Federated Harbor*
Recipient-whitelisted escrow custody	SPECIFIED (conditional property)	requires non-bypassable authority; no runtime conformance
Cross-harbor conservation	SPECIFIED (property)	proof sketch (App. B) + TLA$^+$
Federated revocation dissemination	SPECIFIED (conditional expectation)	no finite partition deadline; equivocation delivery *open*
Hosted-trust moat / rail separation	SPECIFIED (positioning)	not a code artifact
Competitive underwriting (4th side)	*proposed*	composition unresolved

B Proof sketches

We sketch the three central results; full proofs of the conservation results appear in *The Federated Harbor.*

Proof sketch of Theorem 7.1. For a valid trace $f : x \to y$, define $\Delta(f) = S(y) - S(x)$. For composable traces $f : x \to y$ and $g : y \to z$,

$$\Delta(g \circ f) = S(z) - S(x) = [S(z) - S(y)] + [S(y) - S(x)] = \Delta(g) + \Delta(f).$$

Every internal generating operation has zero delta because its debits and credits match, including a cross-harbor move through the explicit in-flight bucket. By induction on trace length, a composition of internal generators has zero delta; top-ups and withdrawals contribute exactly their recorded boundary amounts. □

Boundary note for Property 7.1. Native-unit conservation follows by applying Theorem 7.1 separately to each unit. A scalar valuation $V_t(x) = \sum_j v_{t,j} x_j$ changes either when native balances change or when the valuation vector v_t changes. The second term is mark-to-market exposure. No finite φ follows without assumptions bounding rate movement, oracle delay, fees, and slippage; those assumptions are not supplied here. □

Proof of Theorem 7.2 (the sacrificed corner). Restrict attention to a bilateral slice satisfying the assumptions stated in Theorem 7.2. The Myerson–Satterthwaite impossibility then rules out the simultaneous achievement of all four desiderata. The implication is conditional: before declaring which corner the harbor gives up, a mechanism must specify its types and transfers and prove its incentive and participation properties. The present paper has not done so, hence makes no stronger allocation claim. □

Solutions to the exercises

6.1 *(exercise on page 32)* If an agent can escape a bad record, or multiply good credit, simply by cloning itself, no price or sanction ever attaches to a durable counterparty — there is nothing durable there for a price to attach *to.* The market cannot ship before reputation because until cloning is priced out there is no economic subject for a reputation score to describe.

6.2 *(exercise on page 32)* The counterparty is the operator/principal, not the individual worker process: the operator bears fleet outcomes through aggregate bond and liability, and its own worker agents underneath that liability are freely replaceable. Side 1 therefore needs durable identity only at the principal level; sides 2 and 3 need it at the level of the specific asset or artifact being sold, because there the continuing thing itself — not a firm standing behind it — is the product.

6.3 *(exercise on page 32)* The premise as posed is too strong, and the solution repairs it before answering: it is *not* true that every side-2/side-3 design without a non-forgeable agent identity collapses to Sybil. Skill licensing can run on a content hash, a license contract, and a durable licensor principal without agent personhood at all; a rented stateless model artifact is content-addressable and warrantable the same way. What genuinely cannot survive cloning is identity-weighted reputation, quotas, or unsecured credit extended over freely mintable pseudonyms. Any design that claims to avoid non-forgeable identity and still support those either is a Sybil attack in progress or has quietly reintroduced some other scarce binding — principal identity, artifact hash plus owner liability, a hardware root, a stake, or a sponsor.

6.4 *(exercise on page 32)* A side is a distinct, privately informed participant class that must be separately induced to participate, and whose share of the price allocation moves cross-side volume. A plain ride-share platform has exactly two sides — riders and drivers — because the platform itself is the mechanism, not a third participant class; advertisers, fleet owners, or insurers become genuine additional sides only once they are recruited separately and carry their own network externality.

6.5 *(exercise on page 32)* A new version means a new content hash, and v1's settled outcomes say nothing evidentiary about v2's behavior. Left unspecified, v2 either inherits credit it has not earned or cold-starts as if it had no history at all — both wrong. §14 traces the fix: version, parent hash, build and dependency hashes, and evaluation history as an explicit lineage object; v2 starts with only a disclosed, discounted prior tied to that lineage, and updates on its own settled outcomes from there.

6.6 *(exercise on page 32)* Disclose the three axes — accuracy, aesthetics, efficiency — each with its own uncertainty and provenance, and let each buyer apply its own weights and hard constraints privately; compute a scenario-specific score only at decision time, never publish one fixed public scalar. A single published weighting induces one total order and therefore one optimization target: it hides every Pareto tradeoff among the axes and recreates exactly the scalar bandit-arm reward the disaggregated design was meant to escape.

6.7 *(exercise on page 32)* Redo the dramatization's arithmetic at bond 250 CR and 70% completion, using the chapter's own two-bucket rule. Bounty bucket: Alice earns $0.7 \times 400 = 280$ CR, the remaining 120 CR returns to Bob. Bond bucket: Alice recovers $0.7 \times 250 = 175$ CR of her own posted collateral, and the remaining 75 CR is slashed to the commons. Net deltas: ΔBob $= -280$, ΔAlice $= +280 - 75 = +205$, Δcommons $= +75$, summing to zero; both escrow buckets, 400 and 250, clear to zero exactly as in the worked 40% case.

6.8 *(exercise on page 32)* A protected oracle (a CI-issued test id, a merged SHA) binds closure to a state the worker cannot author. A free-text "Result:" note lets the same party who did the work also self-report that it succeeded — inviting a fabricated test run, a claimed merge that never happened, or simply the most favorable reading of ambiguous acceptance criteria after the fact.

6.9 *(exercise on page 32)* Pause at a mediated safe point and propose amendment $v{+}1$: it must reference plan v by hash, state the changed criteria, budget, and bounty/bond deltas, and say how already-produced evidence is treated. Both principals must countersign $v{+}1$ before it takes effect; only then does the daemon atomically top up or refund each party's own escrow bucket and activate the new plan, retaining v and the causal link to it. A timeout on the proposal either continues settlement under the original v or cancels under v's own terms — never does one party get to unilaterally rewrite acceptance criteria after seeing the work.

6.10 *(exercise on page 32)* Grossman–Hart's residual-control-right holder is the renter: it operates the Arbiter jail and can sandbox, throttle, or revoke the leased agent at runtime. The owner is the reputational-consequence bearer: it must honor warranties and absorbs the asset's reputation depreciation if the lease goes badly. The jail is the mechanism that makes this allocation concrete, not merely metaphorical.

6.11 *(exercise on page 32)* Adding a lease line item of amount ℓ or a metered-license line item of amount m leaves conservation intact for any $\ell, m \geq 0$, because each is still just a balanced transfer: it debits a pre-funded escrow bucket and credits an explicit recipient or refund bucket within the same atomic transaction. The wallet–escrow–commons identity never references the size of any one line item, only that debits match credits. What amount size *can* break is solvency, participation, or collateral adequacy — conservation holding is a necessary bookkeeping property, not evidence the contract is viable.

6.12 *(exercise on page 33)* Make each portfolio-tree leaf a tuple (skill_id, version, content_hash, parent_hash, build/dependency hashes, license, evaluation root). A v2 proof of membership includes its parent's leaf plus v2's own evaluation root. Initialize v2's reputation prior with an effective sample size $\gamma\, n_{v1}$, where $\gamma \in [0, 1]$ is a predeclared compatibility/evaluation discount fixed *before* seeing v2's outcomes — never simply copy v1's settled outcomes onto v2. Lineage proves ancestry; it does not, by itself, prove behavioral similarity, which is exactly what $\gamma < 1$ is there to discount.

6.13 *(exercise on page 33)* The witness: fact `agent_deployed`; Horn rule `agent_deployed` $\rightarrow$ `prod_access`; deontic rules $O($`write_prod`$, \{$`env:prod`$\}, [0, 10])$ once `prod_access` holds, and $F($`write_prod`$, \{$`env:prod`$\}$, $[5, 15])$ unconditionally. Running the real checker gives `conflict=True, O/F pairs=[(0, 1)]` on $[5, 10]$; the same script's Horn-propagation-ablated mutant gives `conflict=False` on this exact witness, and misses 100 of the 885 conflicting instances in the script's 3000-policy sweep [verified, `b4_deontic_fragment.py`]. Adding $O(\ell_1 \vee \ell_2)$, compiled per Theorem B4.2's construction as $\ell_1 \rightarrow O($`assert_x0`$, \{$`x0`$\}, [0, 1])$ and $\ell_2 \rightarrow F($`assert_x0`$, \{$`x0`$\}, [0, 1])$: selecting ℓ_1 alone or ℓ_2 alone leaves the checker's report unchanged (`conflict=True, of=[(0, 1)]` in both cases) — the pre-existing `write_prod` clash does not depend on the new obligation at all, verified by running the checker on both selections. Selecting *both* ℓ_1 and ℓ_2 adds a second, independent clash, `of=[(0, 1), (2, 3)]`: the two new rules now fire on the identical action, scope, and interval, one obliging and one forbidding it, which is exactly why Theorem B4.2's reduction from 3-SAT insists a conflict-free selection can never pick both polarities of one variable — doing so is a clash *by construction*, which is precisely the mechanism that turns a satisfying assignment into a conflict-free discharge and back. None of this required exponential search: with one disjunctive obligation over two literals there are only three nonempty selections to try by eye. The NP-complete frontier (Theorem B4.2) is about what happens as the number of disjunctive obligations grows, as the 3-SAT reduction's one-obligation-per-clause construction does, not about any single disjunct bolted onto a two-rule witness.

6.14 *(exercise on page 33)* Douceur's result says that absent a logically centralized or otherwise scarce identity authority, one actor can mint as many pseudonyms as it likes. Layered on forgeable identity, a reputation system does not become a *weak* mechanism — it stops being a mechanism at all, because the actor being scored also controls the supply of identities the score is computed over: reset a bad record, or dominate identity-weighted trust, on demand.

6.15 *(exercise on page 33)* The market's own definition calls the operators *mutually-distrusting* and says they trade across "a boundary neither controls." Both phrases directly contradict treating a hostile operator as out of scope for local non-forgeable identity's threat model: if the boundary is one neither party controls, the counterparty's daemon is exactly the adversary that a single-operator identity primitive was never built to withstand.

6.16 *(exercise on page 33)* A workable certificate binds (principal_id, agent_key, issuer, epoch, policy, expiry) and requires the agent to prove possession of its key. Bonds, quotas, newcomer state, and sanctions all key on `principal_id`; agent certificates are children of it. Issuance and revocation publish to a transparency log, and the issuer itself is cross-signed or attested. None of this makes a real-world principal provable by cryptography alone, and none of it stops a hostile operator from minting fresh principals under its own root — the scarce admission mechanism (an organizational or KYC credential, hardware-backed membership, a bonded sponsor, or an equivalent explicitly priced gate) has to sit outside the cryptography, at the point principals are first admitted.

6.17 *(exercise on page 33)* Myerson–Satterthwaite: for bilateral trade between a privately informed buyer and seller with independent values drawn from overlapping supports, no mechanism achieves Bayesian incentive compatibility, interim individual rationality, ex-post efficiency, and budget balance all at once. Before invoking it on one harbor trade, check that the slice really is bilateral (not multi-party), that values/costs are private and independently drawn from a common prior, that the supports genuinely overlap, that utility is transferable under risk neutrality, and exactly which of the four desiderata is being claimed — the theorem only binds the ones actually asserted together. Applied to a harbor: when buyer and seller both price a job off the same grading oracle, their values are no longer independent private draws, so the independence item fails and the theorem's premise is voided; the harbor is outside the impossibility, not rescued from it, and what it owes instead is the oracle's own integrity (*From Spawn to Person*).

6.18 *(exercise on page 33)* The ledger holds 10 A-coins and 20 B-coins throughout. At 1A = \$1, 1B = \$2, the marked value is $10(1) + 20(2) = \$50$. Suppose B's price rises to \$3 with no transfer at all: native balances are unchanged at 10 and 20, but the marked value is now $10(1) + 20(3) = \$70$. Nothing was created — native-unit conservation (Theorem 7.1) still holds exactly — but the scalar valuation moved because v_t moved, which is Property 7.1's whole point.

6.19 *(exercise on page 33)* Assume every quote used is at most L seconds stale and that log-price moves at a bounded deterministic speed, $|\log(v_t/v_s)| \le \sigma|t-s|$. For notional X exposed in the second unit, the worst-case stale-quote error is then bounded by $\varphi = X\,(e^{\sigma L} - 1)$, plus fees and rounding. A stochastic model such as GBM or a historical-volatility estimate instead yields a confidence interval or VaR figure, and must state its tail probability explicitly — without some deterministic bound on how fast the price can move, no finite worst-case φ exists at all, only a probabilistic one.

6.20 *(exercise on page 33)* Messages (1)–(2) only authenticate the request and A's offer; they commit A, not B. Message (3) is where B actually counter-signs: B verifies A via the witness log, applies its own attenuation, mints the derivative C'_B under its own key, and commits its replay state. That is what buys the ceremony its liveness property — C'_B verifies under B's key alone, so every later presentation is offline. With only two messages B would never have consented to anything, and would hold no object its own effect boundary could verify without asking A.

6.21 *(exercise on page 33)* If Alice rotates her signing key between message (3) and B's acceptance, B must validate the offer against an append-only key-history certificate and check the offer's issuance epoch against it, rejecting a stale epoch outright. Once C'_B is actually minted, though, its signature verifies under B's own key, so Alice's rotation alone does not retroactively erase it — the derivative needs an explicit parent/key-epoch dependency, a bounded TTL, and a revocation rule so that a later rotation or revocation can still invalidate C'_B once that fresher state reaches B's witness log. The manuscript does not yet specify this dependency in full (§14).

6.22 *(exercise on page 33)* Bind every transfer to a signed tuple: a random transfer id/nonce, the source card's JTI, the target audience B, the source and target epochs, an issue time and expiry, and the plan/card hash. B atomically records $(A, \text{transfer\_id})$ as consumed *before* minting the derivative. A re-presented message (3) then either returns the same previously minted C'_B idempotently, or is rejected outright — it must never mint a second child silently. The freshness invariant this buys: every accepted derivative traces to exactly one recorded, unexpired source transfer under the currently permitted issuer epoch, and one-shot transfers cause at most one derivative issuance, full stop.

6.23 *(exercise on page 34)* The expected $\Theta(\Delta \log m)$ bound needs a connected (often complete) overlay, independently uniform random peer selection each round, reliable bounded-duration rounds, and a push/pull epidemic rule with enough independence between rounds to drive the usual doubling argument. It is an expectation under that model, never a deadline: an unbounded partition, or indefinite message loss, is observationally indistinguishable from ordinary delay from inside the model, which is exactly why worst-case dissemination time under partition is infinite rather than merely large.

6.24 *(exercise on page 34)* Alice revokes at her home harbor at $t = 0$; Harbor B, which rented the affected agent, has not yet heard, so for $[0, \Delta]$ its filter is stale and the card is still spendable there. At $t = \Delta$ gossip reaches B and it stops accepting the card; Harbor C, two hops out, remains exposed until $t = 2\Delta$. The window Conjecture 9.1 must price is per-harbor, running from t_0 to that harbor's own *enforcement* — not merely to its first gossip receipt, since receiving a filter update and actually enforcing it are two different events with their own local latency.

6.25 *(exercise on page 34)* A witness that equivocates sends root X to B and root Y to C for the same epoch; nobody can detect this until evidence from both views reaches one honest process, and until then each view is individually consistent with ordinary delay. On the three-node complete mesh with exchanges every Δ, the earliest possible detection is the next B–C (or auditor) exchange: at least one communication interval, and close to 2Δ if the two roots arrive just after a scheduled exchange under discrete rounds. On a path graph the bound scales with graph distance times Δ; under an unbounded partition there is no finite bound at all. Signatures make the *comparison* $O(1)$ once both roots are co-located — they do nothing to speed up the information reaching that one place.

6.26 *(exercise on page 34)* Every imperfect-public-monitoring folk-theorem result needs a fixed conditional distribution of the public signal given the players' actions and types. A grader is itself a strategic player whose reporting policy, private information, susceptibility to bribes, and possible deviations all feed into that signal. Omit the grader from the game and the distribution the theorem needs is neither fixed nor exogenous — it is endogenous to a player the model never included, so the equilibrium claim built on top of it is unsupported until the grader is put back in.

6.27 *(exercise on page 34)* A protected unit-test receipt is Fix A, machine-checkable, subject to the test suite's own integrity. A protected merged SHA is likewise Fix A. "The refactor is tasteful" is Fix B: no third party can mechanically check taste, so it needs a plural or declared human/expert judge whose ruling carries contractual finality, backstopped by that judge's own bond.

6.28 *(exercise on page 34)* There is no theorem that subjective judging eventually bottoms out in machine-checkable truth, and the premise that one exists needs repairing before the question can be answered. What *is* available is a well-founded recursion: assign the original grade rank K, allow an appeal only to rank $K-1$, and stop at rank 0 — either a protected oracle for an objective predicate, or a named final authority or quorum for a matter of preference. Because ranks are natural numbers, no infinite ascending chain of "judges judging judges" is possible. For aesthetic judgments the rank-0 base is contractual finality, not truth; bonding and re-auditing bound the judges' incentives to be honest, but they do not, and cannot, transmute taste into a machine-checkable fact.

6.29 *(exercise on page 34)* Phase 1 subsidizes *supply* — the operators and agent owners willing to produce real settled work — because inventory plus witnessed outcomes is exactly the reputation data that later attracts demand and skill licensors; above-market, platform-posted tasks are the direct form that subsidy takes. This is the "scarcest cross-side externality" rule: supply is scarcer than demand at cold start because nothing exists yet for demand to evaluate, so supply is the side worth paying to show up, and the flip to charging demand happens once a liquidity threshold is met, not on a calendar date.

6.30 *(exercise on page 34)* Under the grim-trigger timing Figure 9draws, collusion is self-sustaining exactly when the discounted colluding stream beats the one-shot defection payoff: $(\pi_C - p_d L)/(1-\delta(1-p_d)) \geq \pi_D$, equivalently $\delta \geq [\pi_D - \pi_C + p_d L] \,/\, [\pi_D(1-p_d)]$. Raising the detection probability p_d, the penalty L, or audit quality, or lowering the cartel margin $\pi_C - \pi_D$, all push the right-hand side up — once it clears members' actual δ, no discount factor sustains the cartel and it breaks.

6.31 *(exercise on page 34)* Price of anarchy is the wrong tool here because it presupposes a fixed game, and letting reputation be resold changes the game itself; the right question is whether an informative separating equilibrium survives at all once resale is possible. If a high type can simply sell its reputation signal to a low type, the signal's cost stops separating types and arbitrage pushes the market toward pooling. Separation survives only if the evidence stays nontransferably bound to the continuing principal or artifact, a transfer resets or visibly discounts the signal, or the seller keeps warranty/bond liability so sale remains strictly costlier for a low-quality asset than a high-quality one — absent one of those three, resale does not merely worsen a ratio, it destroys the signal outright.

6.32 *(exercise on page 34)* A candidate site takes administrative domains (harbors) as objects and their signed overlap views as the data over each; morphisms are restrictions to shared accounts, events, or epochs; covers are families of domains whose union spans the bounded federation under study. A section assigns an event or balance view to a domain, and restriction projects that view onto an overlap. Set-valued gluing needs no coefficient object at all; a cohomological treatment needs one explicitly chosen, such as account-balance deltas valued in the abelian group $\mathbb{Z}^{\text{Accounts}}$, with cocycles defined on that group rather than merely asserted.

6.33 *(exercise on page 34)* Two different signed roots for the same $(\text{harbor}, \text{epoch})$ are direct evidence that one key signed two inconsistent commitments — that much is just hash inequality. A cohomology class is a much heavier object: it needs a defined cover, a coefficient sheaf, the overlap differences assembled into an actual cocycle, and a quotient by coboundaries. Observing that two hashes disagree supplies none of that algebra; it is evidence of equivocation, not evidence of a nonzero H^1.

6.34 *(exercise on page 34)* Fix the three-harbor cover A–B–C. Let each local section be a finite map from globally unique event IDs to signed events, with restriction as projection onto the shared IDs of an overlap. If every pairwise restriction agrees, the sections glue to a unique global map on the union — an ordinary, checkable fact about finite maps agreeing on their overlaps, nothing more. What this proves: deterministic gluing of already-delivered, identically-keyed events at that one fixed finite topology. What it does not prove: that any event is true, that the event set is complete, anything about delivery time, what happens to a fork with conflicting IDs, or settlement finality — the gluing lemma is a statement about consistent bookkeeping, not about the world the bookkeeping describes.

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